PHOENIX PHASE CONVERTERS

Master Troubleshooting & Wiring Manual

The complete owner's guide to installing, wiring, operating, and troubleshooting every Phoenix Phase Converter model.

All Series: NL · PL · NLT · NLTA · PLT · PLTA · NH · PH · GPX · DualZone · Whisper

Every Symptom · Every Fix · Every Wiring Diagram

Scan-to-Watch Video Guides · Real Product Photos

Support: 800-417-6568 · Text: 602-962-8859

phoenixphaseconverters.com · hub.phoenixphaseconverters.com

Edition 2026.09 · Made in USA · Lifetime Warranty

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Safety First

> Phase converters operate at 230V or 460V — lethal voltages. Before you open a panel, touch a terminal, or begin any diagnostic step in this manual, read this page.

⚠ DANGER — LETHAL VOLTAGE

- Always disconnect power at the main breaker *and* lock it out before opening the converter panel.
- Verify power is off with a multimeter **before touching any conductor**. Trust the meter, not the switch.
- Capacitors store lethal energy **even with power off**. Discharge every capacitor with an insulated bleed resistor or a rated discharge tool before handling.
- Never touch bare terminals with the converter running. The idler generates 240V or 480V continuously while spinning.
- Never spin the idler by hand with power connected. Spin only with the disconnect open and locked out.

Personal protective equipment

☐ Class 0 (1000V) insulated rubber gloves for any panel work
☐ Safety glasses or a full face shield during capacitor discharge
☐ Non-conductive footwear on a dry surface
☐ Remove watches, rings, and any conductive jewelry
☐ Have a second person present when working live

Tools you should have on hand

☐ **True-RMS multimeter** — required for voltage imbalance measurement
☐ **Clamp meter** — for measuring startup inrush and per-leg amp draw
☐ **Capacitance meter** — to test start and run capacitors
☐ **Torque screwdriver** — to torque terminal connections to spec (loose terminals are the #1 cause of intermittent failures)
☐ **Insulated screwdrivers** rated for the working voltage
☐ **Bleed resistor** or capacitor discharge tool
☐ **Megger** (insulation resistance tester) — for tripping-breaker diagnostics

Call a licensed electrician if…

- You are not comfortable working inside an energized electrical panel.
- Your installation requires modifying your main service panel or feeder.
- The converter shows visible burn damage or the smell of burning insulation.
- Any diagnostic step in this manual assumes electrical experience you do not have.

> **When in doubt, call us first.**
> Phoenix Phase Converters support: **800-417-6568** · text **602-962-8859**
> Monday–Friday 7 AM–5 PM MST · Saturday 7 AM–12 PM MST. We answer the phone.

How a Rotary Phase Converter Works

> Understanding what your converter is doing under the hood makes every troubleshooting step in this manual make sense.

The three parts of every Phoenix converter

1. **Single-phase input** — two hot legs (L1, L2) from your utility service, typically 230V.
2. **Idler motor** — a three-phase motor spinning free with no mechanical load. Single-phase input energizes two of its windings; the third winding, driven by the rotating magnetic field, *generates* the missing third phase through electromagnetic induction.
3. **Control panel** — start capacitors kick the idler into rotation, run capacitors balance the three legs during operation, contactors switch loads, and a voltmeter monitors output.

Why the third leg is different

The two "utility" legs (L1 and L2 passed through to T1 and T2) come from your single-phase service. The third leg (T3, sometimes called the "generated leg" or "wild leg") is *manufactured* by the idler motor. It will read slightly different from T1 and T2 under load — this is normal within ±2 % on a properly sized Phoenix unit. When T3 drifts more than 5 % under load, the converter is undersized or has a capacitor problem. This is the single most important number in this manual.

Static vs Rotary vs Digital — where Phoenix models fit

Type	How it works	Voltage balance	Phoenix line
Static	Capacitors + start relay generate a momentary third phase only during startup, then drop out.	Poor (60–70 %)	RD Series
Rotary	Idler motor + control panel produces continuous, balanced three-phase power.	Excellent (±2 %)	NL, PL, NLT, NLTA, PLT, PLTA, NH, PH, DualZone, Whisper
Digital	IGBT power electronics + DSP synthesize a mathematically precise third phase.	Near-perfect (±1 %)	GPX Series

> **Rule of thumb:** for motor loads, a Phoenix rotary is the right choice. For CNC machines, digital inverter welders, or anything with sensitive electronics where every millivolt matters, step up to PL, GPX, or add a line reactor.

Series Guide

Every Phoenix Phase Converter fits into one of these series. Knowing yours is the first step in every troubleshooting flow.

NL Series

Standard 230V rotary. Auto-start when power is applied — no start/stop button on the front lid.

GP3NL — GP100NL · 13 models · 2–100 HP

Input: 230V single-phase · Output: 230V three-phase

PL Series

Premium rotary with front-panel start/stop button and built-in contactor. Enhanced filtering; recommended for CNC and sensitive electronics.

GP3PL — GP100PL · 13 models

Input: 230V single-phase · Output: 230V three-phase

NLT Series

Rotary + integrated step-up transformer. One unit takes 230V single-phase in and delivers 460V three-phase out.

GP3NLT — GP100NLT · 13 models

Input: 230V single-phase · Output: 460V three-phase

NLTA Series

Industrial three-phase with AutoLink protection and auto-balanced 460V output. Adds transformer protection over NLT.

GP3NLTA — GP100NLTA · 13 models

Input: 230V single-phase · Output: 460V three-phase (protected)

PLT / PLTA Series

PL rotary + transformer package. PLT uses an external fusable disconnect; PLTA integrates AutoLink protection with the transformer.

Sized to match every PL model

Input: 230V single-phase · Output: 460V three-phase

NH / PH Series

High-voltage direct rotaries. Take 460V single-phase in from your service and deliver 460V three-phase out — no transformer needed.

Sized 2–100 HP

Input: 460V single-phase · Output: 460V three-phase

GPX Digital Series

Digital rotary with WiFi/cloud energy monitor, 7-day scheduling, remote on/off, and the cleanest output in the lineup. Best for TIG inverters, CNC plasma, precision electronics.

Full HP range

Input: 230V single-phase · Output: 230V and/or 460V three-phase

DualZone Auto-Zoned

Two idler motors in one oversized NEMA 4 / IP66 panel. Maximum power for large shops running multiple high-HP machines or single loads 40 HP+.

GP20/40NL, GP25/50NL, GP30/60NL, GP40/80NL, GP50/100NL · 40–200+ HP

Input: 230V single-phase · Output: 230V three-phase

Whisper Series

Sound-attenuated for noise-sensitive shops, offices, or residential-adjacent installations. Quiet operation under 60 dB.

6 models

Input: 230V single-phase · Output: 230V three-phase

How to tell which series you have

1. **Check the nameplate** — a metal plate on the panel or motor lists the model number.
2. **Look at the front lid** — if there's a green start button and red stop button, it's a PL, PLT, PLTA, PH, or GPX. If the lid is blank, it's an NL, NLT, NLTA, NH, or DualZone.
3. **Count the idler motors** — one idler is standard; two idlers under one panel is a DualZone.
4. **Check output voltage** at T1–T2, T2–T3, T1–T3. 230V ± readings = 230V-out series. 460V ± = high-voltage or transformer-package series.

> **Not sure?** Text a photo of your nameplate to **602-962-8859**. We will confirm the exact model and route you to the right pages in this manual.

Wiring Handbook — All Series

> The next 21 pages contain the official Phoenix Phase Converters wiring diagrams. Every series is covered end-to-end. Use these diagrams as the authoritative reference during installation and any wiring troubleshooting call.

⚠ Before wiring anything

- Disconnect and lock out utility power at the main service panel.
- Verify the converter is sized for your largest load and total connected HP.
- Torque every terminal to the spec printed inside the panel — loose connections cause 30 %+ of "won't start" calls.
- Ground the panel enclosure and the idler frame to your service ground.
- Never share the input breaker with any other load.

How to read a Phoenix wiring diagram

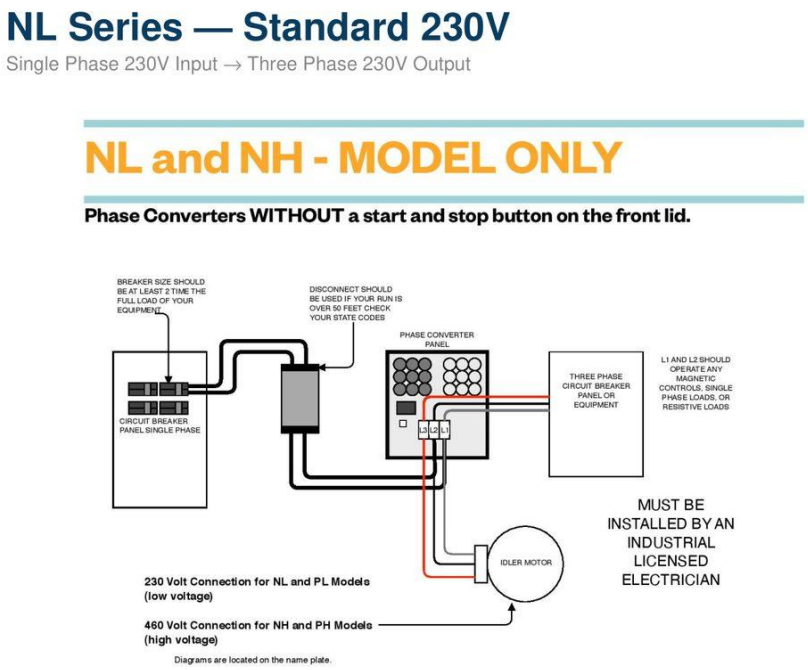
- **L1, L2** — single-phase hot legs from your utility service.
- **L3** — on some diagrams, the internal generated leg reference before it becomes T3.
- **T1, T2, T3** — the three-phase output terminals that feed your equipment. T3 is the generated leg.
- **Ground** — always bonded to the panel enclosure and idler motor frame.
- **Start capacitors** — high-microfarad, in-circuit only during motor start.
- **Run capacitors** — lower-microfarad, always in circuit while running; they balance the third leg voltage.
- **Contactor** — the switch that connects power to the idler when you press start (PL/PH/GPX) or when input power arrives (NL/NH/DualZone).
- **Voltage relay** — protects the converter by dropping the contactor if the incoming voltage is too low.

Each diagram on the following pages is reproduced from the current Phoenix wiring handbook. If your unit was manufactured before the current handbook revision and its physical layout differs from these diagrams, call **800-417-6568** — we keep the older revisions on file and will email you the correct one.

Wiring Diagram — NL & NH Model — Connection Overview

Handbook page 3

High-level connection overview for the standard NL Series (230V→230V) and NH Series (460V→460V) rotary phase converters. Shows L1/L2 input, capacitor bank, contactor path, and T1/T2/T3 output routing to your equipment.



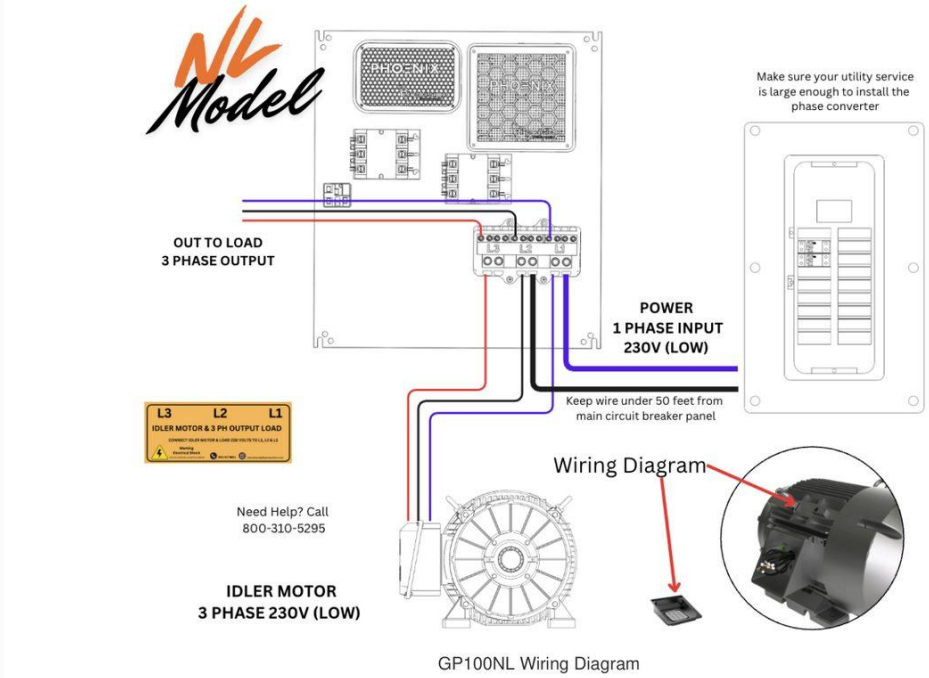
NL & NH Model Diagram

[Error loading GP40NL_Wiring.png: cannot identify image file 'source/GP40NL_Wiring.png']

Wiring Diagram — GP100NL — 100 HP Configuration

Handbook page 4

Full internal wiring for the flagship GP100NL. Note the multiple parallel run capacitors used to hold voltage balance across a 100 HP idler under load.

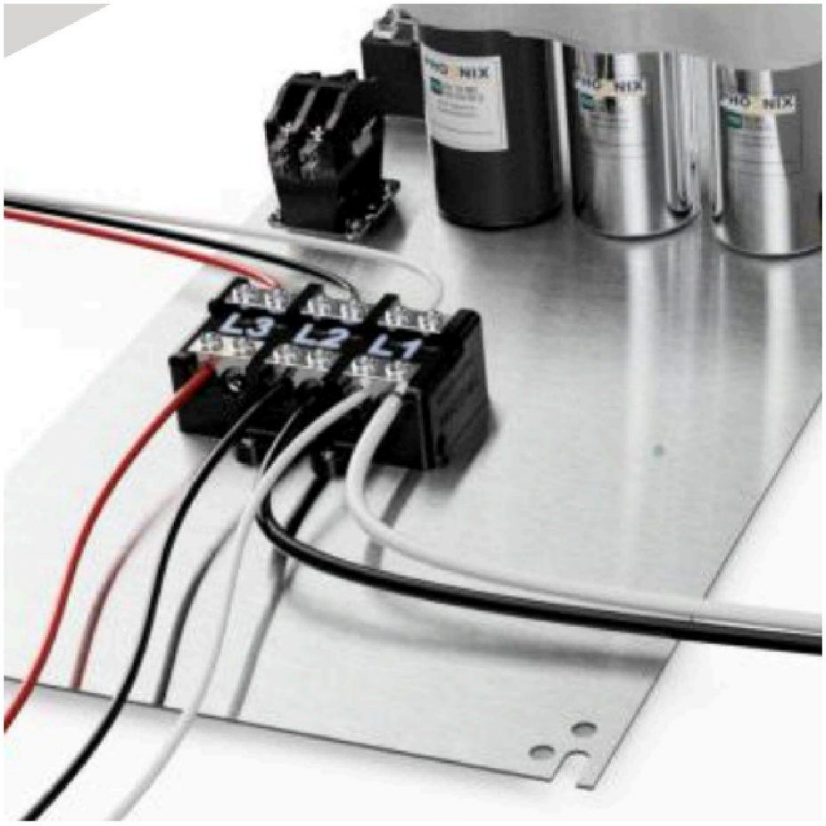


GP100NL Wiring Diagram

Wiring Diagram — NL Model 40 — Terminal Block Connections (L3, L2, L1)

Handbook page 5

Terminal-block detail for the GP40NL. Use this diagram when landing your input feeder wires — pay close attention to the L3 terminal position, which is unique to certain revisions.

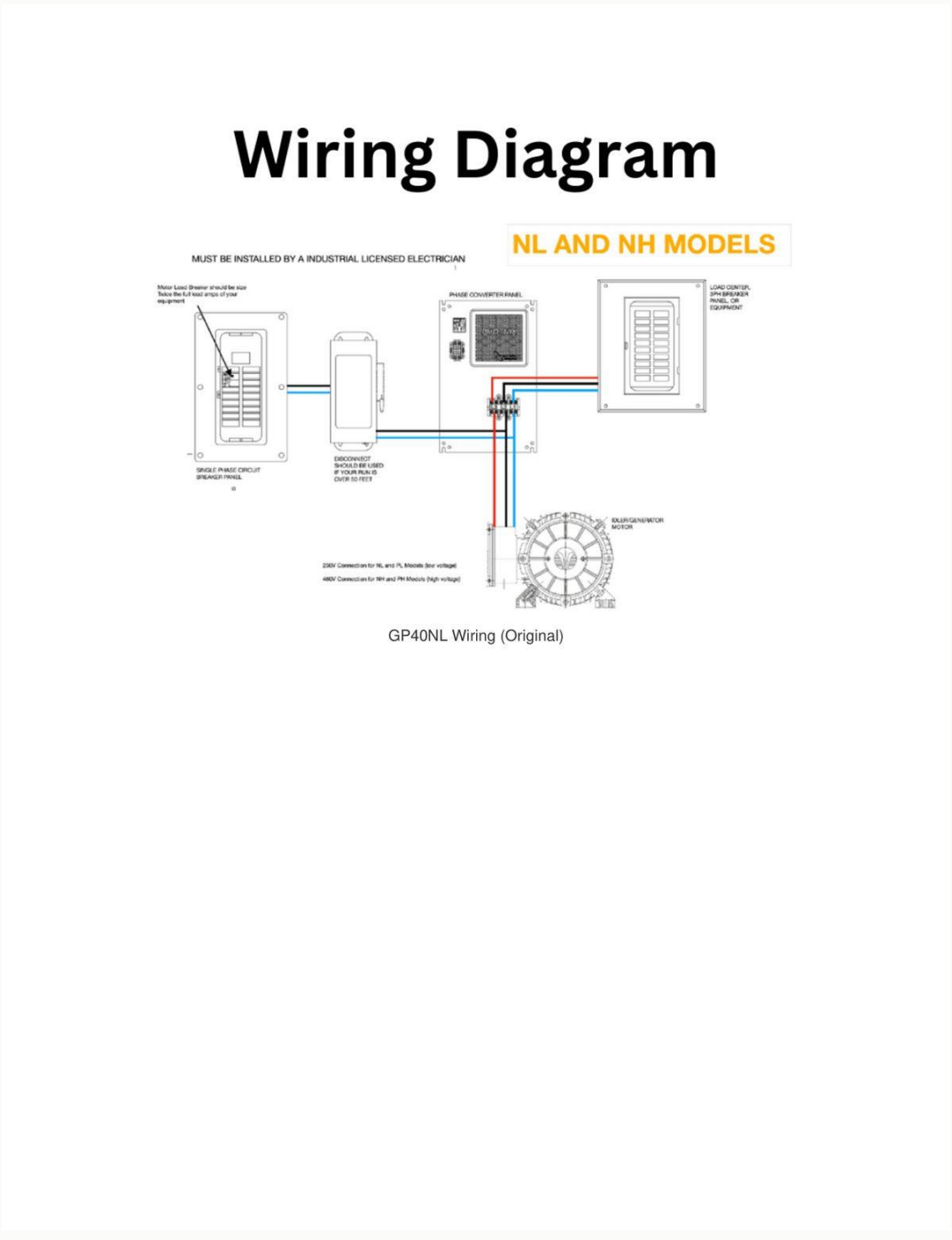


NL Model 40 Connections

Wiring Diagram — GP40NL Original Wiring Diagram

Handbook page 6

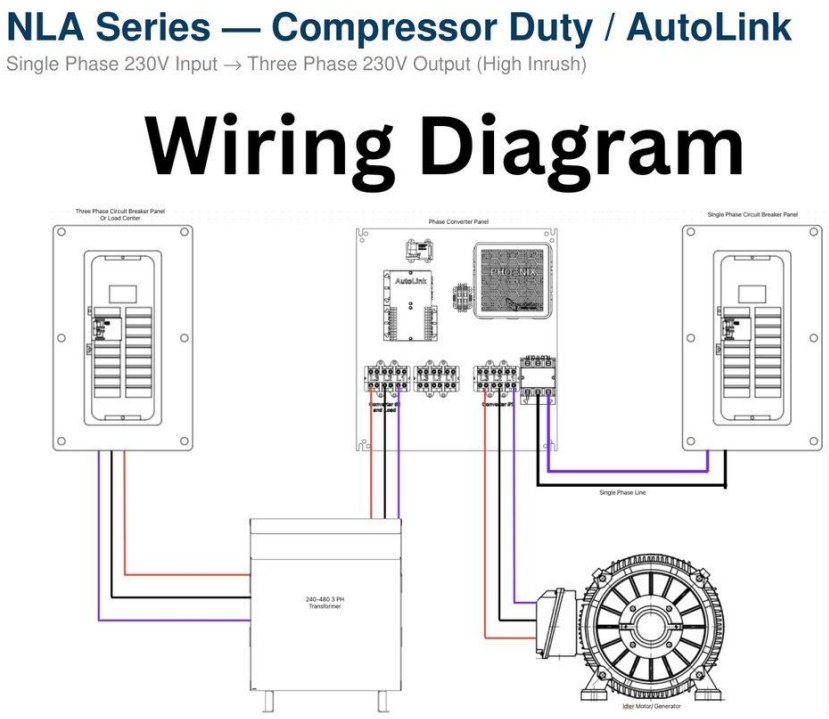
Original-revision GP40NL wiring, applicable to both NL and NH configurations. If your GP40NL was shipped before 2022 and does not match the page 5 layout, use this diagram.



GP40NL Wiring (Original)

Wiring Diagram — NLA / AutoLink — Compressor Duty Configuration

AutoLink wiring for a compressor-duty installation. AutoLink prevents the compressor from restarting simultaneously with the converter after a power interruption — the #1 cause of overcurrent trips on compressor circuits.



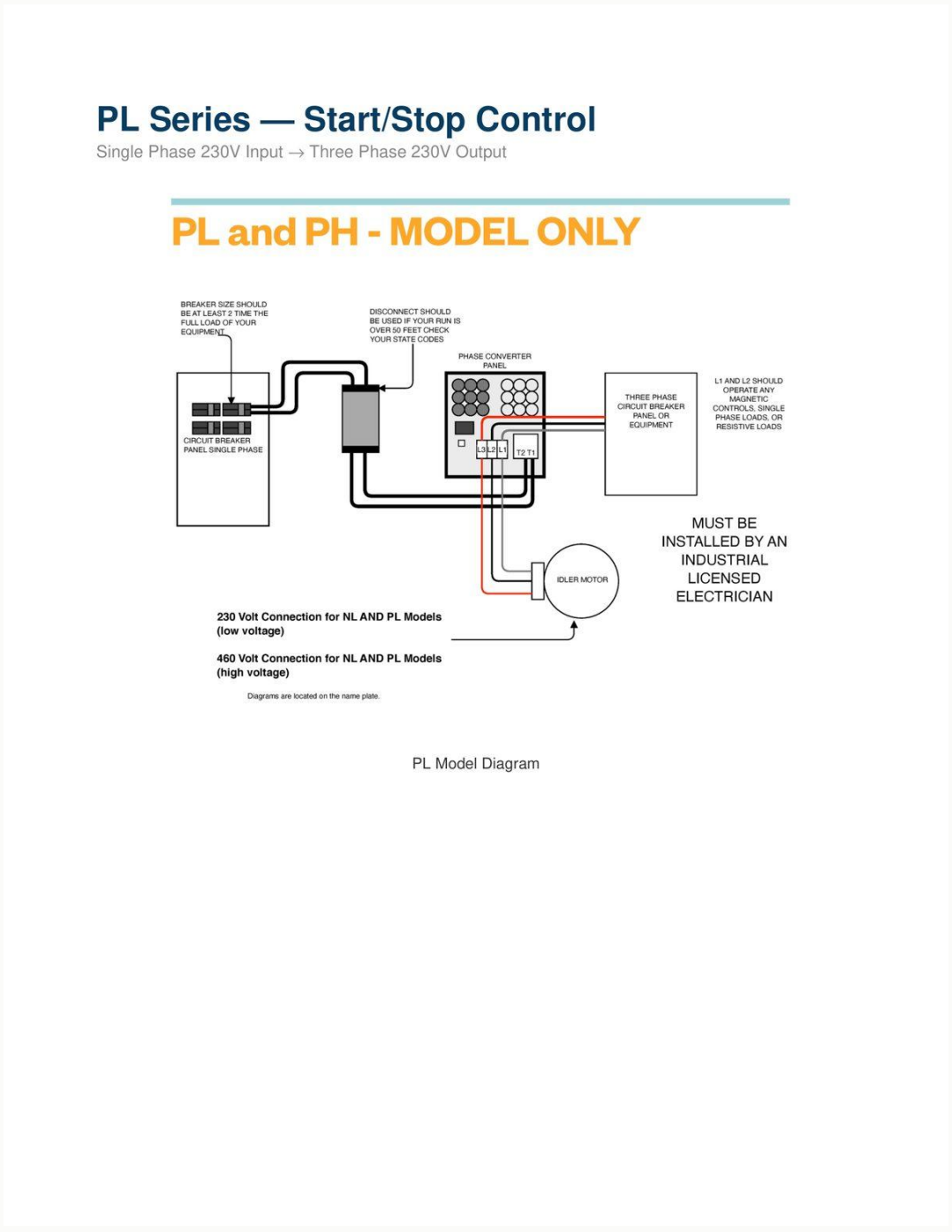
NLA Wiring Diagram

[Error loading NLAL_Wiring.png: cannot identify image file 'source/NLAL_Wiring.png']

Wiring Diagram — PL & PH Model — Connection Overview

Handbook page 8

High-level overview for PL Series (230V→230V, front start/stop) and PH Series (460V→460V, front start/stop). The T1/T2 timer terminals for the start/stop control circuit are shown.

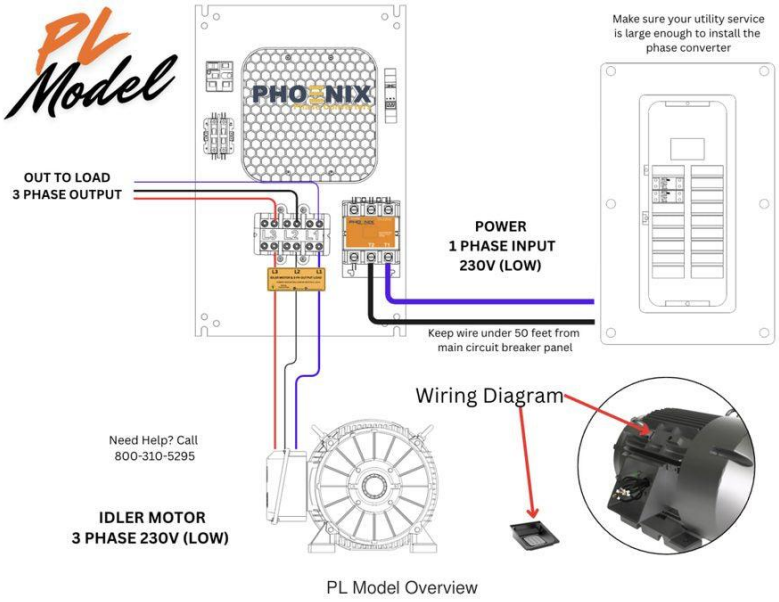


PL Model Diagram

Wiring Diagram — PL Model — Panel Layout and Connections

Handbook page 9

Physical panel layout of a PL rotary. Use this diagram to identify each component during any panel inspection: contactor, start caps, run caps, voltage relay, distribution block, and terminal strip.

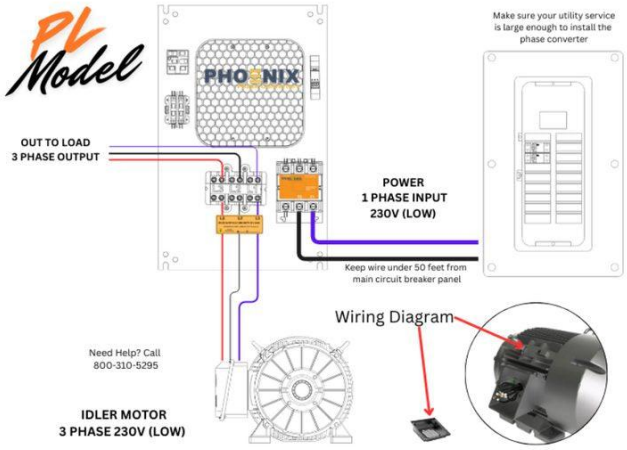


PL Model Overview

Wiring Diagram — PL Model Wiring — Detailed Internal Connections

Handbook page 10

Internal wire routing for the PL series. Every color-coded wire, terminal number, and jumper is shown.

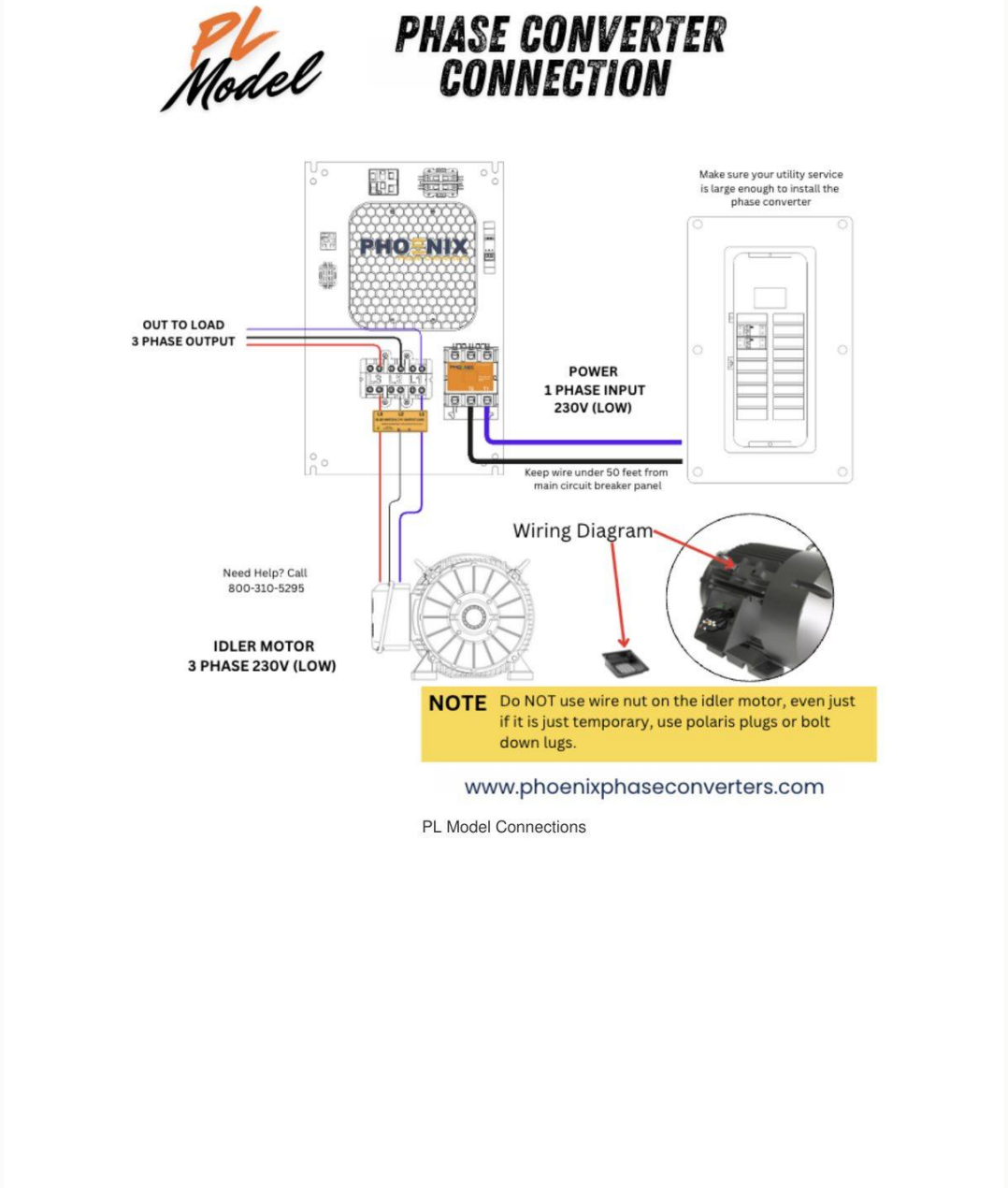


PL Model Wiring

Wiring Diagram — PL Model — Complete Wiring with Notes

Handbook page 11

PL complete wiring including installer notes on wire sizing, torque values, and the small-diameter jumper wire that ties the timer circuit to the contactor coil.

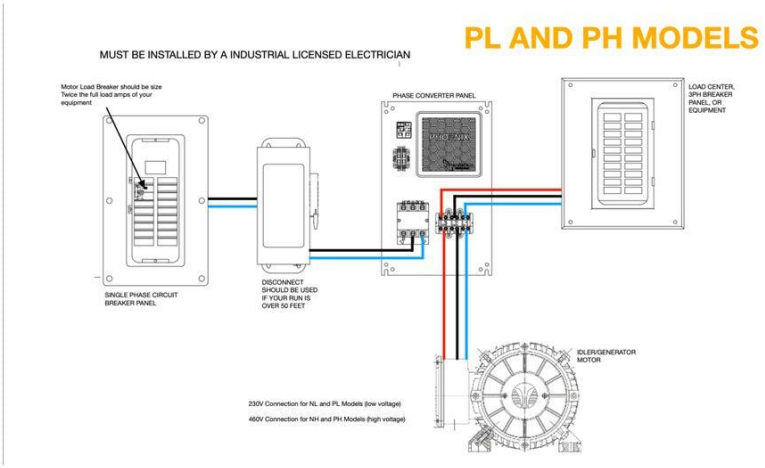


PL Model Connections

Wiring Diagram — PL & PH — Detailed Full Schematic

Handbook page 12

Full electrical schematic (not physical layout) for the PL/PH series. Use this diagram when troubleshooting circuits by function rather than by physical location.



PL & PH Wiring Diagram

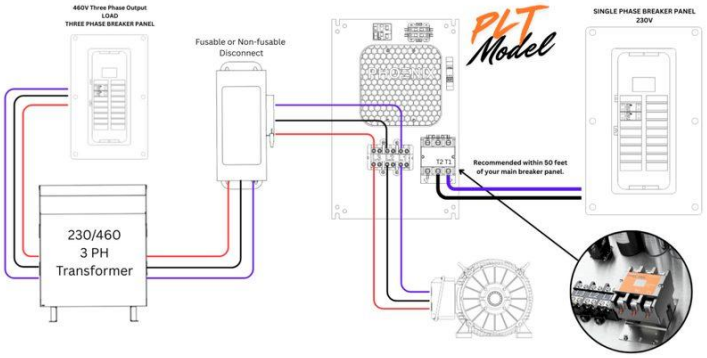
Wiring Diagram — PLT & PLTA — Transformer Configurations

Handbook page 13

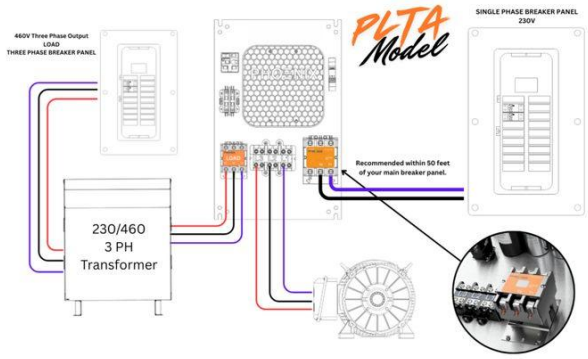
Both transformer-package configurations shown side by side. PLT uses an external fusable disconnect between converter and transformer; PLTA integrates the AutoLink system for lockout protection during power interruptions.

PLT / PLTA Series — 460V Transformer Packages

Single Phase 230V Input → Three Phase 460V Output



PLT Model Diagram

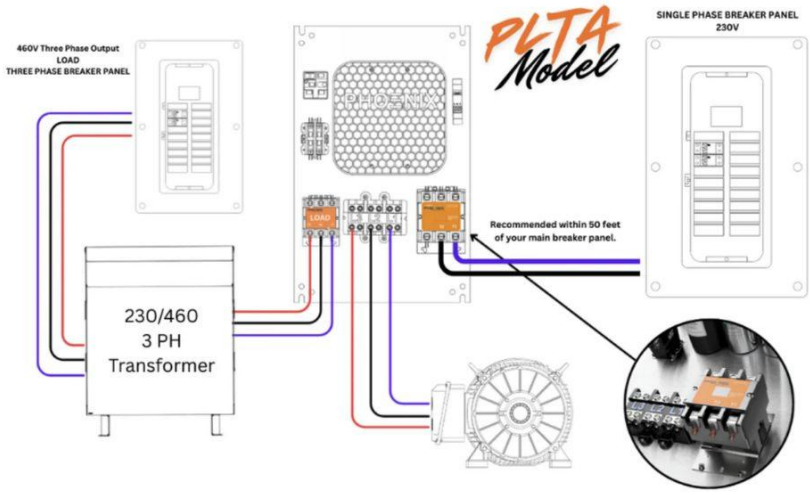


PLTA Wire Diagram

Wiring Diagram — PLTA — AutoLink Transformer Package

Handbook page 14

Complete wiring for a PLTA installation. The AutoLink block sits between the 230V three-phase output of the converter and the 460V input of the step-up transformer, preventing simultaneous re-energization after utility restoration.

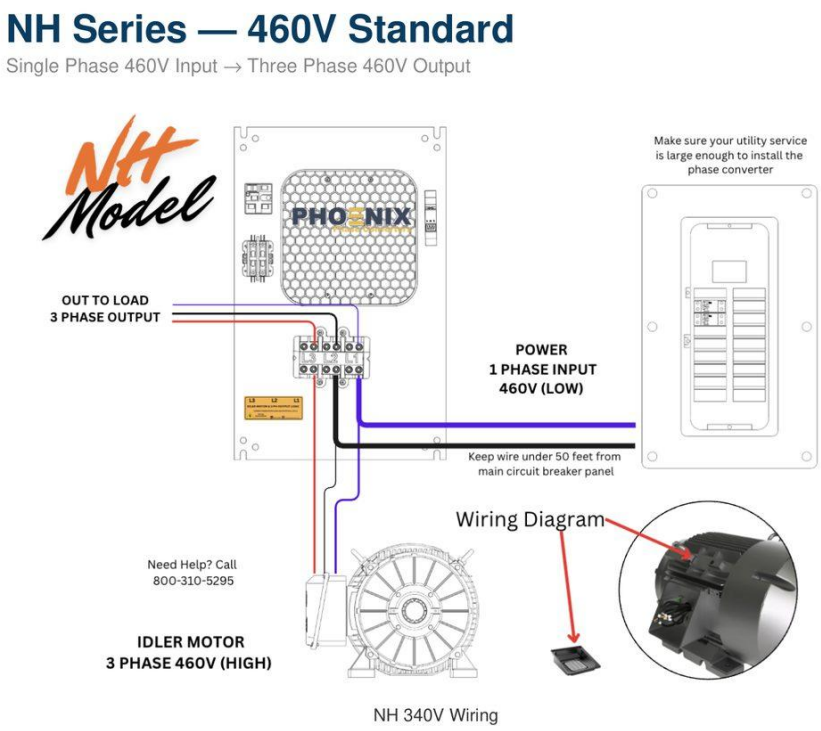


PLTA Wiring Diagram

Wiring Diagram — NH Model 340V Wiring — Panel Layout and Motor Connection

Handbook page 15

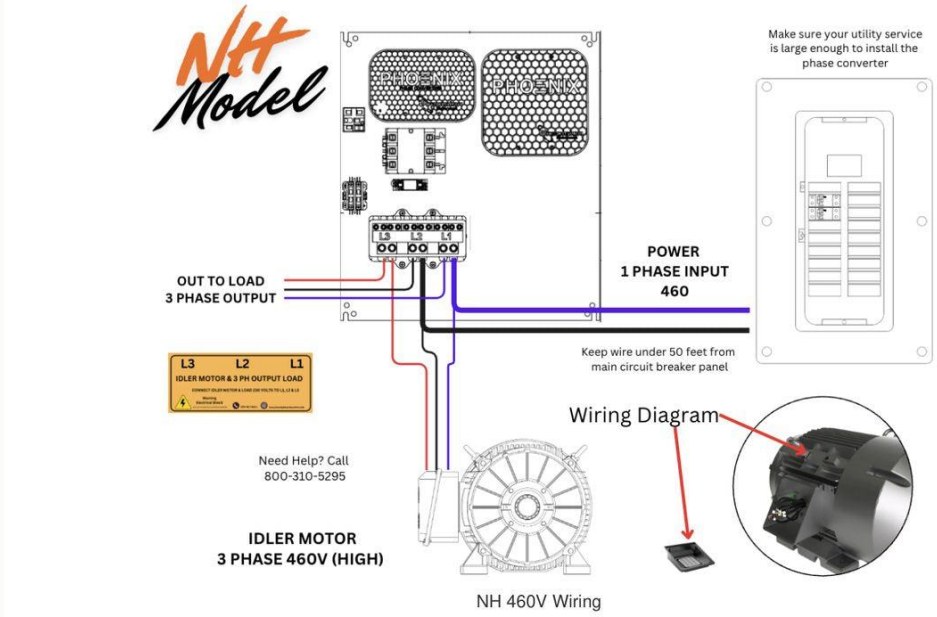
NH panel layout showing the 340V tap arrangement and motor connections. NH-Series converters take a 460V single-phase input; internal jumpers select the correct capacitor voltage rating.



NH 340V Wiring

Wiring Diagram — NH 460V Wiring — Complete High Voltage Configuration

Full NH high-voltage wiring at 460V input. All conductors, capacitors, and contactor coils are rated for the higher operating voltage.

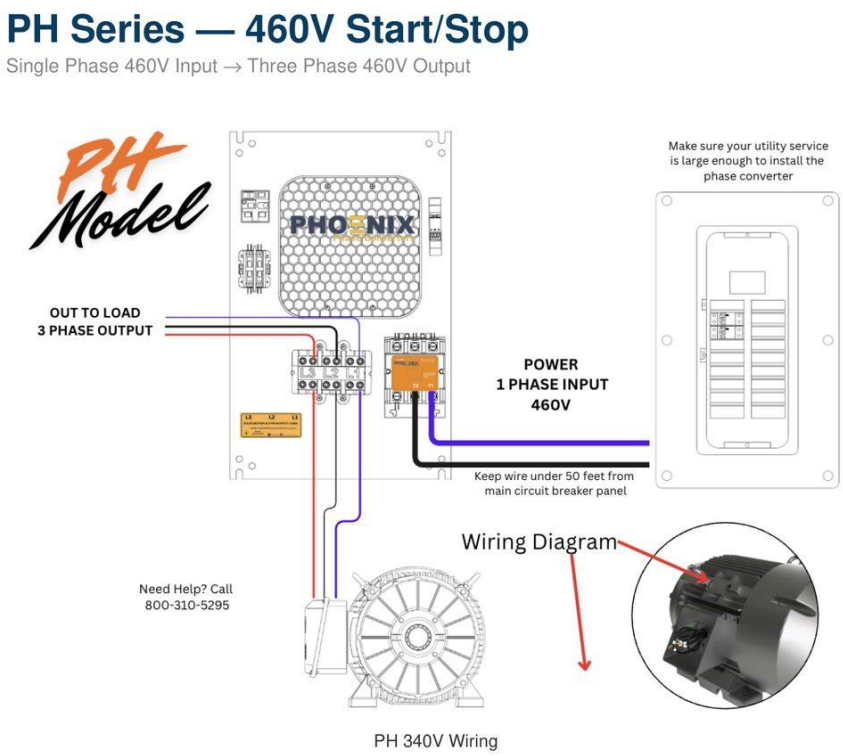


NH 460V Wiring

Wiring Diagram — PH Model 340V Wiring — Panel Layout and Connections

Handbook page 17

PH-series panel layout with the front-panel start/stop button wired into the 340V tap configuration.

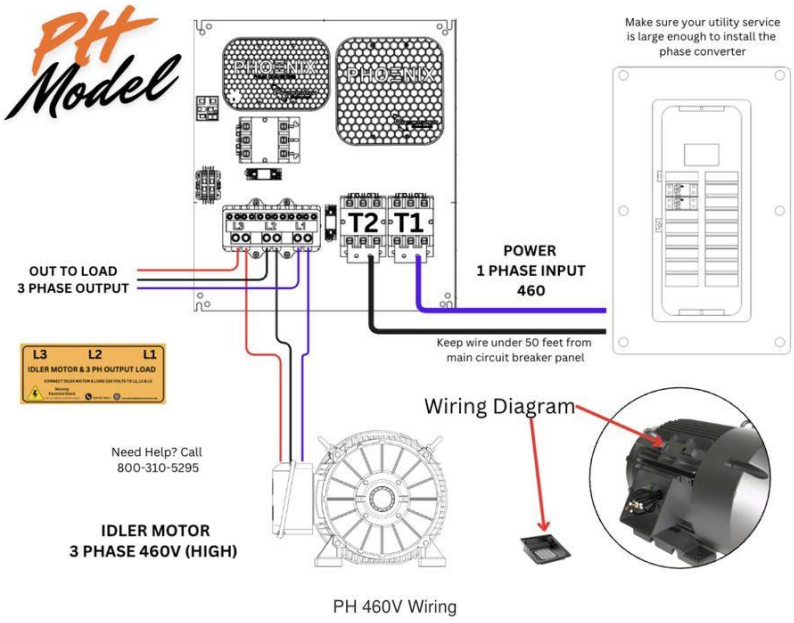


PH 340V Wiring

Wiring Diagram — PH 460V Wiring — Start/Stop Configuration

Handbook page 18

Complete PH high-voltage wiring at 460V input including start/stop control circuit and timer.

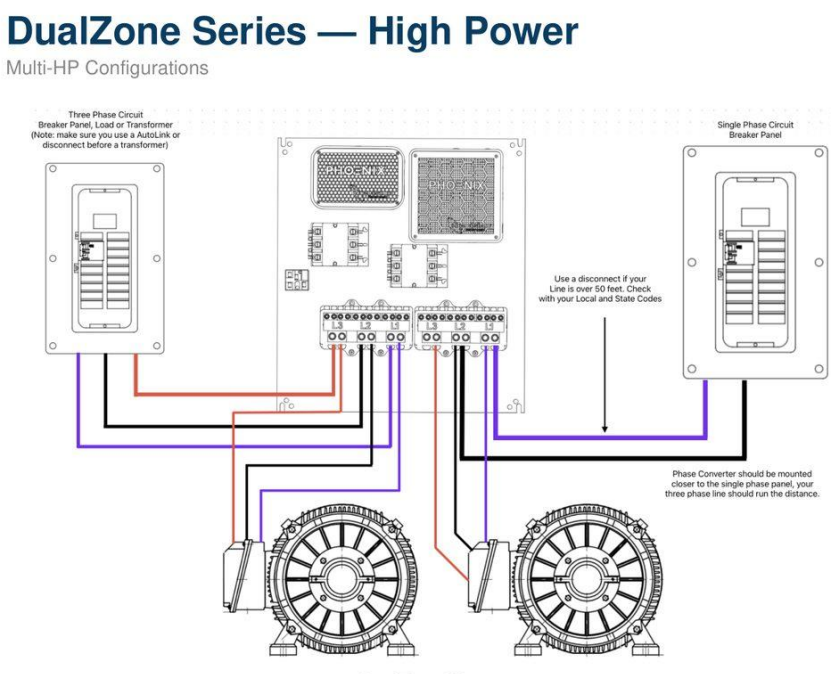


PH 460V Wiring

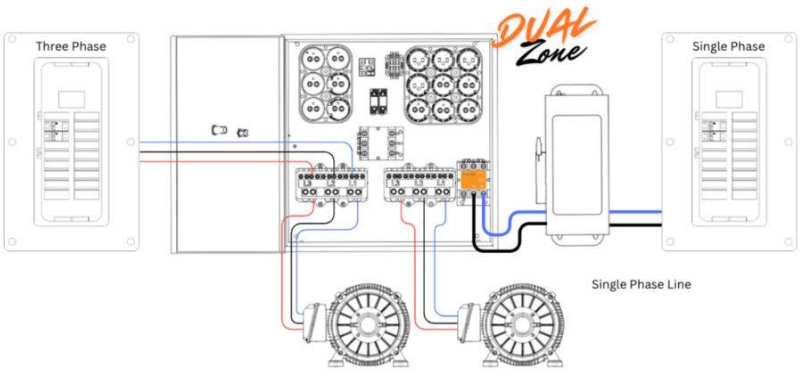
Wiring Diagram — DualZone — Schematic and PL Configuration Wiring

Handbook page 19

DualZone overall electrical schematic combined with the PL-style start/stop control circuit. Shows how the two idler motors share a single control panel while each receives its own dedicated capacitor bank.



DualZone Diagram

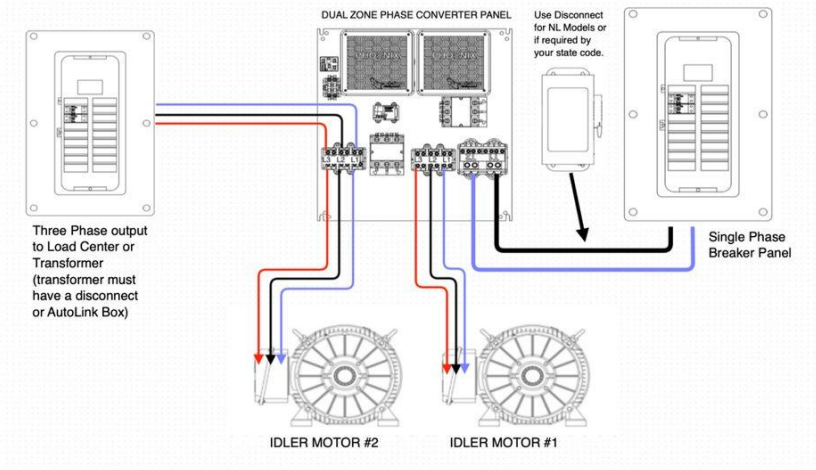


DualZone PL Wiring

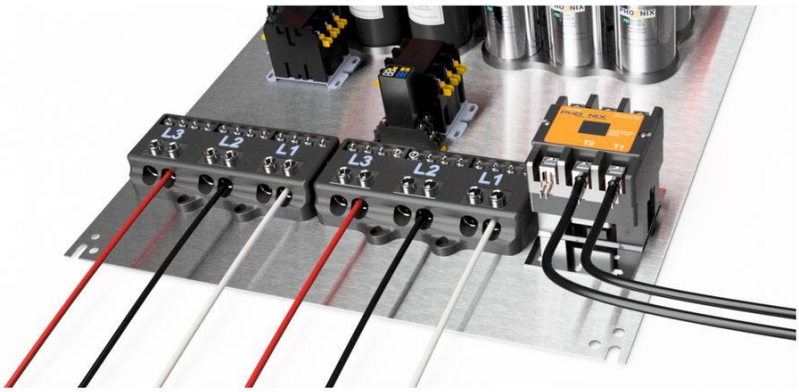
Wiring Diagram — DualZone — Dual Idler Motor Terminal Connections

Handbook page 20

Physical terminal connections for both idler motors in a DualZone unit. Motor 1 handles the base load; Motor 2 auto-cycles in when the current draw exceeds the base-load threshold.



DualZone Wire Diagram



D Diagram

Wiring Diagram — Reference — Motor Cutaway (Three-Phase Idler Internal Components)

Handbook page 21

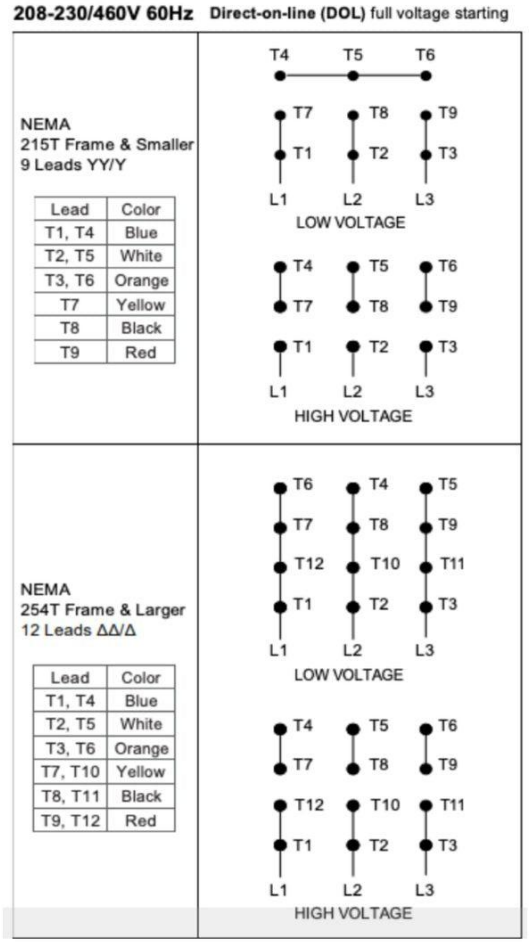
Cutaway of a three-phase idler motor showing rotor, stator, windings, bearings, cooling fan, and terminal box. Reference this diagram anytime you need to identify the internal component causing noise, vibration, or heat.



Wiring Diagram — Reference — 9-Lead and 12-Lead Motor Configurations

Handbook page 22

Standard 9-lead and 12-lead three-phase motor wiring for low-voltage (230V) and high-voltage (460V) connections. This is the diagram you need when landing the T1/T2/T3 output from a Phoenix converter into your three-phase load motor.

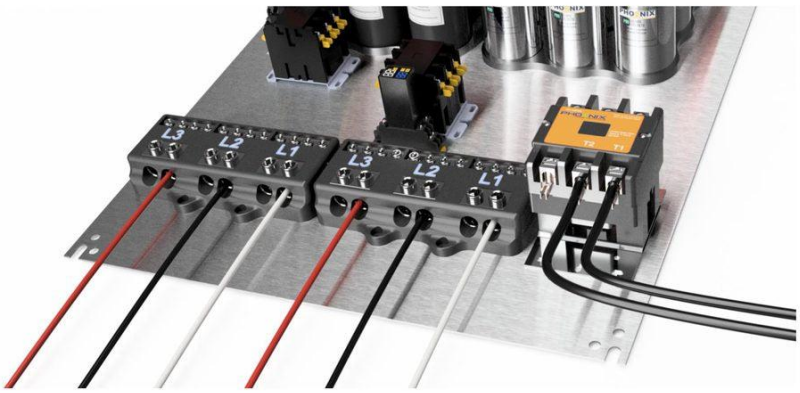


Lead Diagram

Wiring Diagram — Reference — Application Wiring Notes

Handbook page 23

Application-specific wiring notes and common installation configurations. Includes multi-machine shop wiring, sub-panel feeds, and equipment-specific power routing.



Idle Power Draw

Motor Wiring — 9-Lead and 12-Lead

> Most of the "converter is running but my equipment won't power up" calls we receive come down to a motor wired for the wrong voltage. This page is the fix.

Find your motor lead count

Open the motor terminal box. Count the wire leads coming out of the stator:

- **9 leads** — most common on older US industrial motors. Numbered 1 through 9.
- **12 leads** — common on European imports and modern high-efficiency motors. Numbered 1 through 12.
- **6 leads** — dual-voltage but simpler; treat as a 9-lead subset.
- **3 leads** — single-voltage motor; no reconfiguration possible. Match the converter voltage to the motor voltage.

9-lead motor — 230V (low voltage) connection

For a 9-lead motor being fed 230V three-phase from a standard NL, PL, or DualZone converter:

- T1 → tie leads 1 & 7 together, connect to converter T1
- T2 → tie leads 2 & 8 together, connect to converter T2
- T3 → tie leads 3 & 9 together, connect to converter T3
- Tie leads 4, 5, 6 together, insulate — do not connect to any incoming power

9 Wire Leads Motor - Low Voltage 230V

Scan for the step-by-step 9-lead low-voltage wiring video

youtube.com/watch?v=mVtPR4ESg6s

12-lead motor — 230V (low voltage) connection

For a 12-lead motor being fed 230V three-phase:

- Tie leads 1, 7 together → converter T1
- Tie leads 2, 8 together → converter T2
- Tie leads 3, 9 together → converter T3

- Tie leads 4, 5, 6 together
- Tie leads 10, 11, 12 together
- Do NOT connect the two tied groups (4/5/6 and 10/11/12) to each other or to power

12 Lead Motor - Low Voltage 230V

Scan for the step-by-step 12-lead low-voltage wiring video

youtube.com/watch?v=BujkVL6-fG4

How to Connect a 12-Wire Motor

Scan for the alternative 12-wire motor connection walkthrough

youtube.com/watch?v=BJ4zstQH-B8

Reversing motor rotation

If your motor runs in the wrong direction after wiring:

1. Shut off power completely at the disconnect.
2. Verify power is off with a multimeter.
3. At the *motor terminal box* (not the converter), swap any two of the three phase leads.
4. Restore power and confirm rotation.

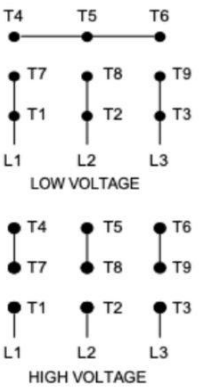
> Swap at the motor, not at the converter output. Swapping at the converter output will reverse rotation for *every* three-phase load downstream — usually not what you want.

Reference diagram

208-230/460V 60Hz **Direct-on-line (DOL)** full voltage starting

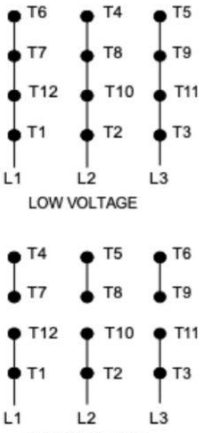
NEMA
215T Frame & Smaller
9 Leads YY/Y

Lead	Color
T1, T4	Blue
T2, T5	White
T3, T6	Orange
T7	Yellow
T8	Black
T9	Red



NEMA
254T Frame & Larger
12 Leads ΔΔ/Δ

Lead	Color
T1, T4	Blue
T2, T5	White
T3, T6	Orange
T7, T10	Yellow
T8, T11	Black
T9, T12	Red



Lead Diagram

Master Troubleshooting Reference

> Every problem we have ever seen, in one place. Each entry follows the same structure: symptoms → likely causes → step-by-step diagnostic procedure → fix → when to call us.

Diagnostic order — always follow this sequence

1. **Safety** — power off, verify off, discharge caps if opening the panel.
2. **Verify utility voltage** — measure incoming L1–L2 at the converter input terminals with the disconnect closed and no load. Should be 230V ±5 % (or 460V ±5 % for NH/PH). Anything under 225V is a utility problem, not a converter problem.
3. **Read the voltmeter** on the panel with the converter running under no load. All three legs should be within ±2 %.
4. **Under load** — recheck all three legs with your largest load running. This is where imbalance and undersizing show up.
5. **Only then** open the panel to inspect components.

Skipping steps 2–4 and going straight to component-level troubleshooting is the #1 way people waste a Saturday afternoon.

1. Converter Won't Start — No Sound

> Every step above except step 1 requires power to be off at the disconnect. Verify with your meter before touching any terminal.

SYMPTOMS

- Push the start button (PL/PH/GPX) or apply power (NL/NH/DualZone) — nothing happens
- No hum, no click, no rotation
- Panel voltmeter reads zero or does not illuminate

LIKELY CAUSES

- No incoming utility power to the converter
- Tripped breaker or blown fuse in the main service panel
- Loose or disconnected input wire at the converter terminal block
- Failed input contactor (PL/PH/GPX only) — no click when start button is pressed
- Failed control transformer (PL/PH/GPX) — no low-voltage supply to the coil
- Failed voltmeter (converter may actually be running — verify with clamp meter)

Diagnostic steps

1. Verify the main breaker feeding the converter is ON and has not tripped. Reset if tripped.
2. Measure voltage at the converter input terminals (L1–L2) with a multimeter. Should read 225–250V (or 450–480V for NH/PH). If zero, the problem is upstream — check panel breaker, fuse, disconnect switch.
3. If input voltage is present but the converter does nothing, check for loose input wires. Torque all input terminals to spec.
4. On PL/PH/GPX: listen for a contactor click when the start button is pressed. No click = failed start button, failed control transformer, or open coil on the contactor.
5. Test the start button for continuity with the panel de-energized.
6. If the contactor coil is not energizing, measure control-circuit voltage at the coil terminals. 120V or 24V (depending on model) should be present when start is pressed.

> **Fix:** Restore incoming power, tighten loose terminals, or replace the failed control component (start button, contactor coil, control transformer). Call us for the correct replacement part number for your model.



Troubleshooting Overview V1

youtube.com/watch?v=wqbkEcqrl_0

Troubleshooting Walkthrough

youtube.com/watch?v=bxn4VsamebQ

2. Converter Hums But Won't Spin

> Start capacitors store lethal energy for hours after power is removed. Always discharge before handling.

SYMPTOMS

- Audible 60 Hz hum from the idler motor
- Rotor does not rotate, or tries to and stalls
- Idler feels hot to the touch after only 30–60 seconds

LIKELY CAUSES

- Failed start capacitor(s) — the #1 cause
- Failed start relay or centrifugal switch
- Seized idler bearings
- Undersized incoming feeder — voltage sags too far during start attempt
- Foreign object jamming the rotor

Diagnostic steps

1. SHUT OFF POWER IMMEDIATELY — a humming, non-rotating motor will burn its windings in under 2 minutes.
2. Discharge all start and run capacitors with an insulated bleed resistor.
3. With power off, try to spin the idler shaft by hand at the fan end. It should rotate smoothly with light resistance. Rough, grinding, or completely locked = bearing failure.
4. If the shaft spins freely, test each start capacitor with a capacitance meter. Compare reading to the µF value printed on the capacitor. Any capacitor more than 10 % below rated value is failed — replace it.
5. Test the start relay for continuity. It should click and close when energized.
6. Measure incoming voltage while a helper attempts to start the converter. If voltage sags below 200V during start, the feeder or utility supply is undersized. Watch the voltage drop video (QR below).

> **Fix:** Replace failed capacitors with the exact-match value and voltage rating specified for your model. Replace failed bearings before restarting — a hum-and-stall converter run repeatedly will destroy its stator windings. Contact us for the capacitor kit for your specific model.

Checking for Voltage Drop

youtube.com/watch?v=gCGoN_nBVZ0

Checking for Voltage Drop V1.0

youtube.com/watch?v=ZszESYXSSwE

Troubleshooting Overview V1

youtube.com/watch?v=wqbkEcqrl_0

3. Growling or Grinding Noise

SYMPTOMS

- Loud growling, grinding, or rumbling from the idler when running
- Noise gets worse under load or as the unit warms up
- Vibration transmitting into the mounting surface

LIKELY CAUSES

- Failing idler bearings — the most common cause
- Voltage drop on incoming supply causing the motor to labor
- Loose mounting bolts allowing frame resonance
- Foreign debris inside the motor cooling fan or between rotor and stator
- Undersized converter overloaded by a load exceeding rated HP

Diagnostic steps

1. Shut off power. Discharge caps.
2. Spin the idler by hand. Roughness, catching, or grinding = bearing replacement needed.
3. Inspect the cooling fan at the non-drive end for debris or damage.
4. Restore power. Under no load, measure voltage on all three output legs. All should be within ±2 %.
5. Bring the largest single load online. Re-measure all three output legs. Growling that appears only under load with T3 dropping more than 5 % from T1/T2 indicates the converter is undersized or has failed run capacitors.
6. Watch the voltage drop video (below) if you suspect utility supply is causing the noise.

Fix: Replace bearings if roughness is present (proactive replacement is far cheaper than a burned-out stator). Tighten mounting hardware and add vibration isolation pads if resonance is the cause. If undersized, upsize the converter or reduce simultaneously running loads.

Checking for Voltage Drop

youtube.com/watch?v=gCGoN_nBVZ0



Technical Support - Wiring Help & Setup

youtube.com/watch?v=kRVblowXkcQ

4. Breaker Trips on Startup

SYMPTOMS

- The input breaker (or a downstream breaker) trips within seconds of power application or the start button being pressed
- Breaker resets normally but trips again on the next start attempt

LIKELY CAUSES

- Input breaker undersized for converter inrush current
- Wrong breaker type (should be a standard thermal-magnetic; some AFCI/GFCI-protected circuits false-trip on motor inrush)
- Simultaneous load — another motor or heavy load starting at the same time as the converter
- Short circuit in converter wiring or a downstream cable
- Failed start capacitor drawing excessive current

Diagnostic steps

1. Verify the breaker is sized correctly for your converter. Rule of thumb: the input breaker should be about 250 % of the converter's rated single-phase input amperage.
2. Confirm the breaker is a standard thermal-magnetic 2-pole type, not an AFCI or GFCI (unless required by local code — in which case a motor-rated version is required).
3. Clamp-measure input current during a start attempt. Compare peak inrush to breaker instantaneous trip rating.
4. Bring the converter online BEFORE any downstream equipment. Never start machines while the converter is spinning up.
5. If a downstream breaker trips, check for a short on that circuit with the converter off. Measure insulation resistance to ground on all three legs with a megger.
6. Verify all capacitor values with a capacitance meter — a shorted capacitor can double the input current.

Fix: Upsize the input breaker if undersized (verify against your feeder wire ampacity first). Replace failed capacitors. Change your operating procedure to always start the converter first, then downstream equipment one machine at a time.

Troubleshooting Overview V1

youtube.com/watch?v=wqbkEcqrl_0

5. Breaker Trips Under Load

SYMPTOMS

- Converter starts and runs fine unloaded
- Breaker trips when a specific machine is brought online, or during heavy cutting/loading
- Trip is delayed — not instant like a short

LIKELY CAUSES

- Converter is undersized for the connected load
- Cumulative load exceeds converter rated capacity
- Wrong load-type sizing multiplier used (very-hard loads like air compressors need ×2.5, not ×1.0)
- Voltage imbalance causing excessive current draw on one leg

Diagnostic steps

1. Note WHEN the trip happens — during motor start = inrush problem; steady-state under cut/load = capacity problem.
2. Clamp-meter all three output legs while the load is running. Compare to converter rated amps.
3. Add up all three-phase loads that run simultaneously. Compare to converter HP rating using the correct load multiplier: easy loads ×1.0, medium ×1.5, hard ×2.0, very hard ×2.5.
4. If a single load exceeds the converter's spindle-motor-start rating, that load cannot be started even with the correct total sizing.
5. Check for voltage imbalance under load — see Issue 6.

> **Fix:** Upsize the converter to match the correct load multiplier, or stagger machine startups so peak inrush never coincides. For CNC machines, always use ×2.0. Call us with your equipment list and we will verify the sizing.

6. Voltage Imbalance Between Phases

SYMPTOMS

- Voltage readings differ by more than 5 % across L1–L2, L2–L3, and L1–L3
- Motors running hot despite being within HP rating
- Equipment performing inconsistently or underperforming
- CNC machines throwing error codes related to power quality

LIKELY CAUSES

- Converter undersized for load (T3 sags most under load)
- Failed or drifted run capacitors
- Loose terminal connections
- Utility supply itself imbalanced or low
- Wrong wire size on the single-phase input causing voltage drop

Diagnostic steps

1. Measure all three legs under load with a true-RMS multimeter. Record L1–L2, L2–L3, L1–L3.
2. Calculate imbalance: (Max − Min) ÷ Average × 100. Anything under 3 % is excellent. 3–5 % is acceptable. 5–10 % is a problem. Over 10 % is critical.
3. Check incoming single-phase voltage at the converter input. It should be 230–250V under load. Below 225V is a utility or feeder problem.
4. Inspect and torque every terminal in the panel. Look for discoloration or scorching — signs of past heating from loose connections.
5. Test every run capacitor. Replace any that read more than 10 % below rated µF.
6. Verify converter sizing. Your unit should be rated for at least the HP of your largest motor plus 20 % headroom.

Fix: Tighten and torque all terminals. Replace drifted capacitors as a matched set (not just one). Upsize the converter or the input feeder if the fundamentals are undersized. For CNC and precision electronics, step up to a PL or GPX series — PL and GPX hold tighter than NL under transient loads.

Troubleshooting Overview V1
youtube.com/watch?v=wqbkEcqrl_0

CNC Precision Balanced Voltage
youtube.com/watch?v=fU0sDAQTuVY

7. Low Voltage on the Generated Leg (T3)

SYMPTOMS

- T1–T2 reads correct (230V ±)
- T3–T1 and T3–T2 read low, often 200V or below under load
- Three-phase equipment struggles, motors hum, contactors chatter

LIKELY CAUSES

- Converter undersized for load — T3 is always the first casualty of undersizing
- Failed or drifted run capacitors — run caps balance the third leg
- Static converter being used where a rotary is needed (T3 is only present during motor start)
- Wrong replacement capacitor value installed after a prior service

Diagnostic steps

1. Confirm you own a rotary converter (Phoenix NL, PL, DualZone, etc.). If you own a static converter, the low T3 under load is expected — upgrade to a rotary.
2. Test every run capacitor. Match the reading to the value printed on the body. Replace any more than 10 % off spec.
3. Verify capacitor values match the factory specification for your model. Substituting the wrong µF value is a common repair-shop mistake and will cause exactly this symptom.
4. Check total connected load versus converter rating. Under-sizing shows here first.
5. If T3 is low even at no load, one or more run capacitors is completely open — this is easy to feel: an open cap runs cold while a healthy one runs warm.

Fix: Replace run capacitors with matched, exact-spec parts. Never substitute across models without confirming the µF and voltage rating. Upsize the converter if sizing is inadequate.

Troubleshooting Overview V1
youtube.com/watch?v=wqbkEcqrl_0

CNC Precision Balanced Voltage
youtube.com/watch?v=fU0sDAQTuVY

8. Overheating

SYMPTOMS

- Idler motor frame hot to the touch (over 140 °F / 60 °C sustained is too hot)
- Thermal breaker or overload relay tripping after extended run
- Burning smell from the panel
- Panel enclosure warm on the outside

LIKELY CAUSES

- Inadequate airflow around the converter (needs 12 in. clearance minimum on all sides)
- Dust and debris blocking cooling fins or the cooling fan
- Converter running consistently above 80 % of rated capacity
- Voltage imbalance forcing one winding to overheat
- High ambient temperature (un-air-conditioned shop in summer)
- Failed cooling fan on the idler motor

Diagnostic steps

1. Verify at least 12 in. clearance on all sides. Move the converter if it is shoved against a wall or crowded by other equipment.
2. With power off and caps discharged, blow out internal dust with compressed air at no more than 30 PSI (higher pressure damages windings).
3. Clean the cooling fan blades at the non-drive end of the idler.
4. Clamp-meter all three legs under normal load. If you are consistently above 80 % of the rated amps, the converter is being pushed too hard.
5. Address any voltage imbalance from Issue 6 — imbalance compounds overheating.
6. Check ambient temperature. Add ventilation or relocate the converter if the ambient exceeds 104 °F / 40 °C.

> **Fix:** Improve airflow, clean dust, address imbalance, or upsize the converter. For hot Arizona shops, positioning the converter under an evaporative cooler or in a shaded corner drops idler temps by 20+ °F.

9. Excessive Noise or Vibration

SYMPTOMS

- Louder-than-normal operation
- Vibration transmitting into floor, wall, or bench
- Rattling from inside the panel

LIKELY CAUSES

- Loose mounting hardware
- Failing bearings (see Issue 3)
- Loose components inside the panel — a capacitor or relay that has come off its mounting
- Missing or worn vibration isolation pads
- Idler frame resonance with the mounting structure

Diagnostic steps

1. Locate the noise source with a mechanic's stethoscope (or a long screwdriver — blade to unit, handle to your ear).
2. Tighten all mounting bolts including through-bolts on the idler.
3. With power off, open the panel and physically check each capacitor, relay, contactor, and wire bundle for looseness.
4. Spin the idler by hand — roughness means bearings.
5. Add rubber vibration isolation pads or a vibration isolation kit if the noise is telegraphing into the structure.

> **Fix:** Retighten, replace bearings, or isolate the mount. Any internal component that has come loose should be secured with lock washers or thread-locking compound before restart.

10. Motor Runs in Wrong Direction

SYMPTOMS

- Everything appears to work, but the driven equipment rotates backward
- Only affects motors — resistive loads (heaters, welders) don't care

LIKELY CAUSES

- Phase sequence at the motor terminals is reversed

Diagnostic steps

1. Shut off power at the disconnect and lock out.
2. Verify power is off with your multimeter.
3. Open the affected motor's terminal box (not the converter).
4. Swap any two of the three phase leads at the motor. Do not swap the ground.
5. Restore power and verify rotation.

> **Fix:** Two-lead swap at the motor. Never swap at the converter output if you have more than one three-phase load — every downstream load will reverse.

11. CNC Won't Power Up (Converter Running)

SYMPTOMS

- Rotary converter is spinning normally
- Voltmeter shows correct three-phase output
- CNC control panel is dark or throwing a power-fault error on startup

LIKELY CAUSES

- Converter is not fully spun up before CNC is energized
- Voltage imbalance beyond CNC input tolerance (typically ±5 %)
- T3 low under load — CNC is more sensitive than a plain motor to power quality
- Machine's internal disconnect not closed, or main power switch off
- Wrong voltage — a 460V CNC connected to a 230V converter, or vice versa
- CNC needs a line reactor for power conditioning

Diagnostic steps

1. Verify all three phase voltages are present and within ±3 % at the CNC's three-phase input terminals, not at the converter.
2. Confirm the CNC's internal disconnect and main power switch are on.
3. Wait for the converter to fully spin up (typically 3–5 seconds after start) before energizing the CNC.
4. Measure T3–T1 and T3–T2 while the CNC power switch is engaged. A drop below 218V (5 %) is CNC-sensitive.
5. If imbalance is greater than 3 % under load, upgrade to a PL or GPX series, or add a three-phase line reactor sized for the CNC's input amps.
6. Verify the converter is sized with a ×2.0 multiplier for the CNC spindle motor — never ×1.0 or ×1.5.

Fix: Sequence: converter first, then CNC. For persistent power-fault errors on CNC despite adequate sizing, add a matched line reactor at the CNC input or upgrade to GPX Digital for near-perfect voltage.

CNC Precision Balanced Voltage

youtube.com/watch?v=fU0sDAQTuVY

Real-World Test - 3 CNC Machines

youtube.com/watch?v=zi9L7Jf3P8o

12. Capacitor Failure — General

> Capacitors store lethal energy for hours after power is removed. Always discharge before handling. Wear safety glasses.

SYMPTOMS

- Swollen, leaking, or discolored capacitor visible in the panel
- Blown-off pressure vent (small button on the top of the capacitor)
- Burning smell traced to a specific capacitor
- Symptoms from Issues 2, 6, or 7 that trace back to a bad cap

LIKELY CAUSES

- Age — capacitors are wear items with typical 5–10 year lifespan under load
- Voltage transient or lightning surge
- Excessive heat in the panel from Issue 8
- Wrong replacement value installed in a prior service
- Manufacturing defect (rare)

Diagnostic steps

1. Shut off power. Wait 10 minutes. Discharge every capacitor with an insulated bleed resistor.
2. Visually inspect all capacitors. Swollen or leaking = replace immediately.
3. Test every capacitor with a capacitance meter. Value on the meter must be within 10 % of the µF printed on the body.
4. Replace failed capacitors with EXACT-MATCH specifications: same µF, same voltage rating, same type (start vs. run).
5. Install a surge protector on the converter input if you are in a lightning-prone area or on rural utility.

> **Fix:** Replace failed capacitors. Use Phoenix-supplied parts to guarantee spec match. Never substitute a start capacitor for a run capacitor or vice versa — they are engineered for different duty cycles.

Troubleshooting Overview V1

youtube.com/watch?v=wqbkEcqrl_0

13. Contactor Issues

SYMPTOMS

- Loud buzzing or chatter from the contactor when it energizes
- Contactor won't pull in when start button is pressed
- Contactor stuck closed — converter won't shut off
- Pitted or welded contacts visible when the contactor is de-energized

LIKELY CAUSES

- Low control voltage — coil not receiving full voltage
- Failed coil — open winding measured across the coil terminals
- Worn or pitted contacts from repeated high-current switching
- Welded contacts from a fault current event
- Debris or contamination in the contactor mechanism

Diagnostic steps

1. Shut off power and discharge caps.
2. Measure coil resistance across the coil terminals. Compare to spec — an open coil reads infinity; a healthy coil reads a specific ohm value.
3. Inspect contact faces visually. Pitting is normal wear; welding or heavy discoloration means replacement.
4. With power on, measure control voltage at the coil while the start button is pressed. Should be 24V or 120V depending on model. Low or missing = control transformer or start button problem.
5. If the contactor is stuck closed, replace immediately — a welded contactor is a safety hazard.

Fix: Replace the contactor with the exact-spec Phoenix replacement part. Contactor life is typically 100,000 cycles under normal load — expect replacement at some point during a converter's lifetime.

Technical Support - Wiring Help & Setup

youtube.com/watch?v=kRVblowXkcQ

14. Voltage Drop on Incoming Supply

SYMPTOMS

- Converter won't start reliably
- Growling noise on start attempt
- Utility voltage measured at the converter reads below 225V when a start attempt is made

LIKELY CAUSES

- Feeder wire from the panel to the converter undersized for the distance
- Loose neutral or feeder connection at the main panel
- Rural utility with marginal service capacity
- Sharing the feeder with another heavy load

Diagnostic steps

1. Measure voltage at the converter input terminals with a multimeter while a helper attempts to start the converter. Watch the reading during the start attempt.
2. If voltage sags below 200V during start, the feeder is undersized OR the utility service is marginal.
3. Measure at the main panel while starting. If the drop is present at the main panel too, the utility supply is the problem — call the utility.
4. If the drop is present at the converter but not at the main panel, the feeder is the problem. Look up wire size for your amperage and distance in the NEC wire-sizing table. Common cause: 100A converter on 60A wire that was fine when the shop was smaller.
5. Check every terminal from the utility drop through the main panel to the converter for tightness and heat discoloration.

Fix: Upsize the feeder to Phoenix-recommended wire size for your converter and distance. For a rural utility supply, add a soft-start kit to reduce inrush by 50–70 %, or contact the utility about upgrading service.



Checking for Voltage Drop

youtube.com/watch?v=gCGoN_nBVZ0



Checking for Voltage Drop V1.0

youtube.com/watch?v=ZszESYXSSwE

15. Environmental & Mounting Issues

SYMPTOMS

- Corrosion or rust on the panel
- Water ingress after a rainstorm
- Bugs, mice, or nesting material inside the enclosure
- Frame cracks or wall pull-out

LIKELY CAUSES

- Wrong enclosure rating for the environment (indoor unit installed outdoors)
- Mounting orientation blocks door seal
- Improperly sealed conduit entry allowing water
- Insufficient wall structure to hold the weight

Diagnostic steps

1. Confirm your enclosure is rated NEMA 4 or IP66 for outdoor. Phoenix NL, PL, and DualZone units come as NEMA 4 by default.
2. Verify the panel is mounted vertically with the door on the front — the gasket does not seal properly in horizontal or angled orientations.
3. Inspect every conduit entry for proper sealant or weather-tight fittings.
4. Check the enclosure gasket for cracks, dry rot, or nesting damage.
5. For pest ingress, seal every unused conduit knockout with a proper cap and consider installing steel-wool packing around cable entries.
6. Verify the wall structure can support the panel weight (see the model spec page).

> **Fix:** Reseal conduits, replace damaged gaskets, and mount properly. If the wrong enclosure was installed for the environment, contact us for a replacement rated for outdoor use.

Application-Specific Guides

> The physics is the same across every application, but the failure modes cluster differently depending on what you are running.

CNC Machines

- **Sizing multiplier:** ×2.0 for the spindle motor HP. Never use ×1.0.
- **Voltage requirement:** Most CNCs need ±3 % or better. Use PL or GPX Digital for anything critical.
- **Line reactor:** Recommended at the CNC input on any converter installation. Provides additional filtering and protects the CNC from transients.
- **Startup sequence:** Always converter first, wait 5 seconds, then CNC power on.
- **Multiple CNCs:** Size for the largest spindle × 2.0 + sum of all other CNCs at ×1.5.

Air Compressors

- **Sizing multiplier:** ×2.5 (very hard load).
- **AutoLink recommended:** Prevents the compressor from restarting simultaneously with the converter after a power interruption — the #1 cause of overcurrent trips on compressor circuits.
- **Duty cycle:** Verify the converter can handle continuous compressor cycling (many shop compressors run 30–50 % duty). Consult the model spec.
- **Two-stage compressors:** Peak current on cold starts can be 6× rated. Size the converter for that peak, not the running amps.

Welders — MIG, TIG, Stick, Plasma

- **MIG 200–250 A:** NL Series, 10–15 HP.
- **MIG 300–400 A:** PL Series, 15–25 HP.
- **TIG 200–350 A (inverter):** GPX Digital preferred, 10–20 HP. Inverter welders are voltage-sensitive.
- **Stick 250–500 A:** PL Series, 15–30 HP.
- **CNC Plasma:** GPX Digital for cleanest power, or PL Series 10–25 HP.
- **Multi-station shops (3–5 stations):** PL Series 30–60 HP, size for the two largest running simultaneously.

HVAC & Rooftop Units

- **Sizing multiplier:** ×2.5.
- **Voltage:** Most commercial RTUs need 460V three-phase. Use NLT, NLTA, PLT, or PLTA for a 230V single-phase service.
- **Multiple compressors:** Never allow all compressors to start simultaneously — stagger with a time delay or AutoLink.
- **Reheat / defrost cycles:** Account for the peak combined draw when auxiliary elements engage.

Elevators & Lifts

- **Sizing:** Elevators require careful sizing — refer to the elevator-specific converter and transformer sizing rules.
- **Regenerative braking:** Some elevator drives regenerate on down-travel. Coordinate with the elevator installer on whether a braking resistor is needed.
- **Code compliance:** Elevator installations often require a licensed electrician and inspector sign-off — do not attempt DIY.

Pumps & Hydraulics

- **Sizing multiplier:** ×2.0 (hard load).
- **Deep-well pumps:** Long lead length to the motor can cause voltage drop. Verify wire size for the well depth.
- **Hydraulic pumps:** Load varies with pressure demand. Size for the pump's max-pressure amp draw, not the rating plate.

Machine Shops with Multiple Machines

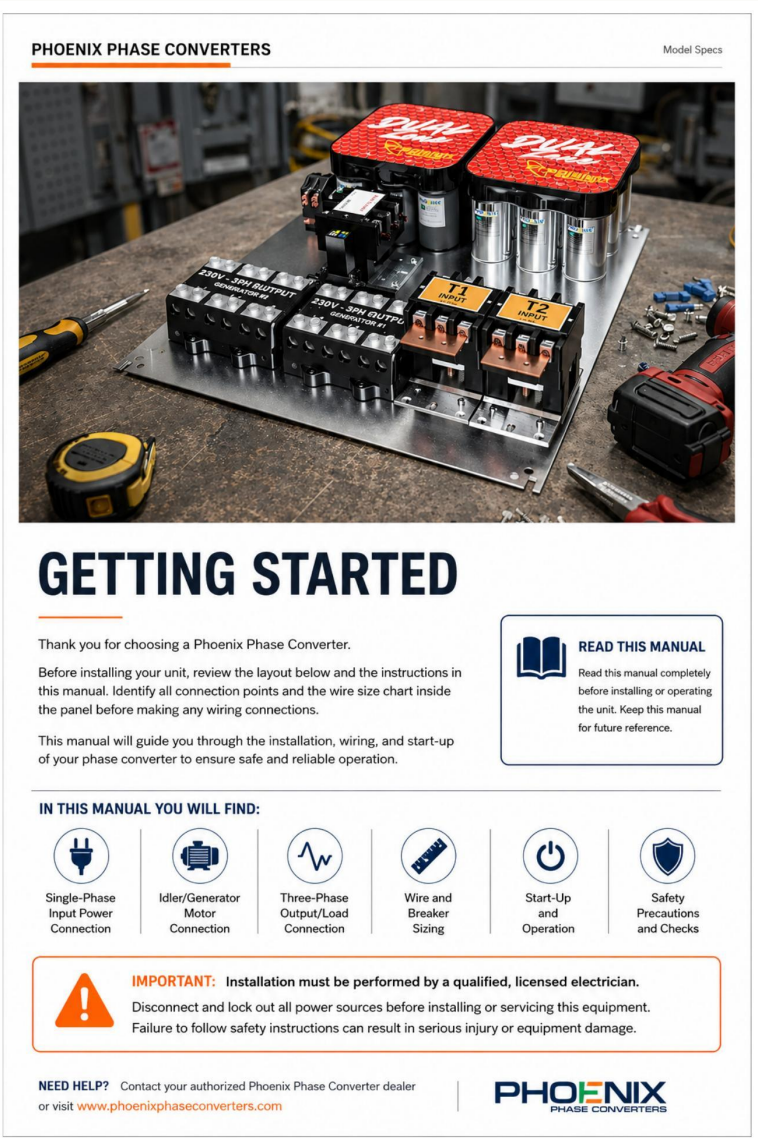
- **Sizing:** Size for the largest single-machine startup at its correct multiplier + sum of all other simultaneously running machines at ×1.0.
- **Sub-panel:** Install a three-phase sub-panel downstream of the converter to feed individual machine breakers.
- **Startup sequence:** Converter first, then equipment one machine at a time.
- **DualZone:** Two-idler DualZone units handle multi-machine shops with less imbalance drift than a single oversized idler.

DualZone Manual

The DualZone Auto-Zoned Power series uses two idler motors in one panel to deliver maximum power to shops running multiple high-HP machines. This section reproduces the full DualZone installation and operation manual.

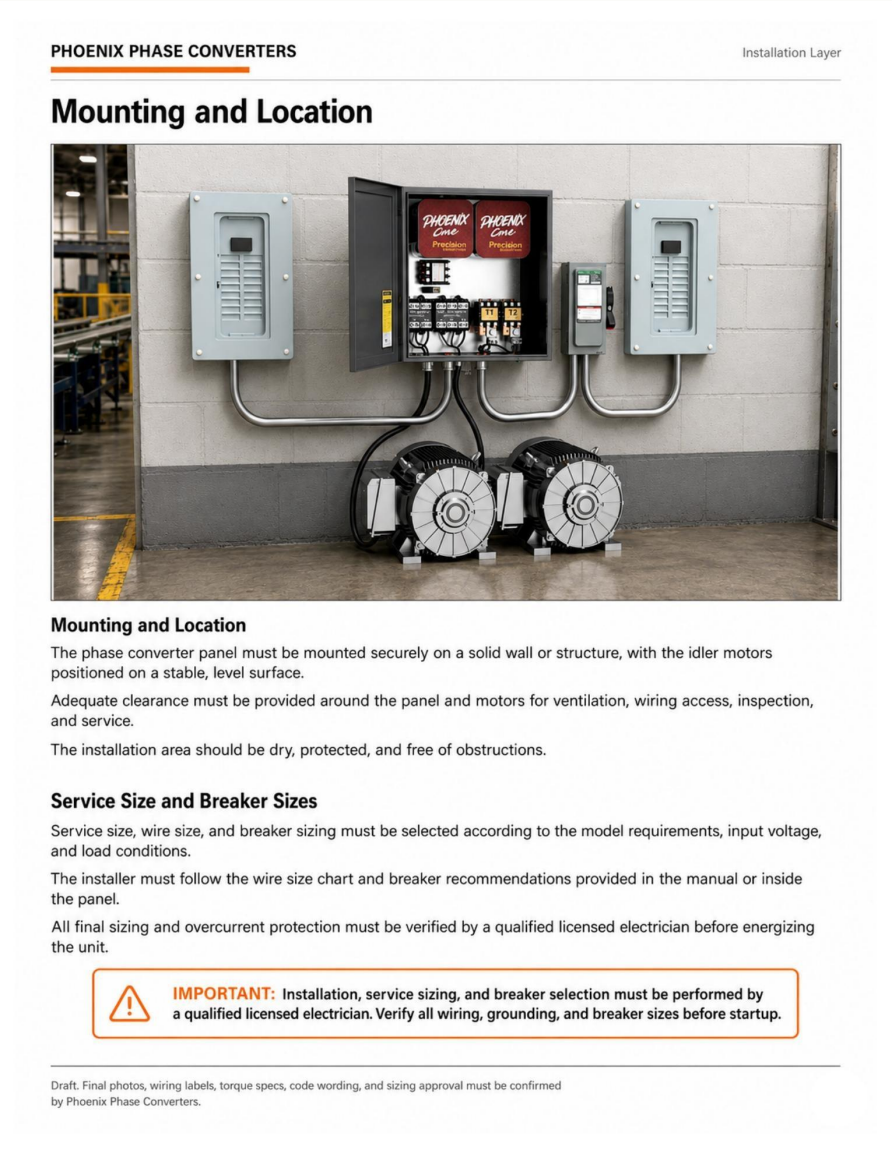


Getting started



Before power is connected, verify the model on the nameplate matches your order and that the enclosure has arrived undamaged. DualZone models are: GP20/40NL, GP25/50NL, GP30/60NL, GP40/80NL, and GP50/100NL.

Mounting location



PHOENIX PHASE CONVERTERS

Installation Layer

Mounting and Location

Mounting and Location

The phase converter panel must be mounted securely on a solid wall or structure, with the idler motors positioned on a stable, level surface.

Adequate clearance must be provided around the panel and motors for ventilation, wiring access, inspection, and service.

The installation area should be dry, protected, and free of obstructions.

Service Size and Breaker Sizes

Service size, wire size, and breaker sizing must be selected according to the model requirements, input voltage, and load conditions.

The installer must follow the wire size chart and breaker recommendations provided in the manual or inside the panel.

All final sizing and overcurrent protection must be verified by a qualified licensed electrician before energizing the unit.

IMPORTANT: Installation, service sizing, and breaker selection must be performed by a qualified licensed electrician. Verify all wiring, grounding, and breaker sizes before startup.

Draft. Final photos, wiring labels, torque specs, code wording, and sizing approval must be confirmed by Phoenix Phase Converters.

- NEMA 4 / IP66 rated — installs indoors or outdoors.
- Minimum 12 in. clearance on all sides for airflow.
- Mount vertically with the door on the front. Never mount horizontally.
- Wall structure must support two idler motors — see the spec page for weights.

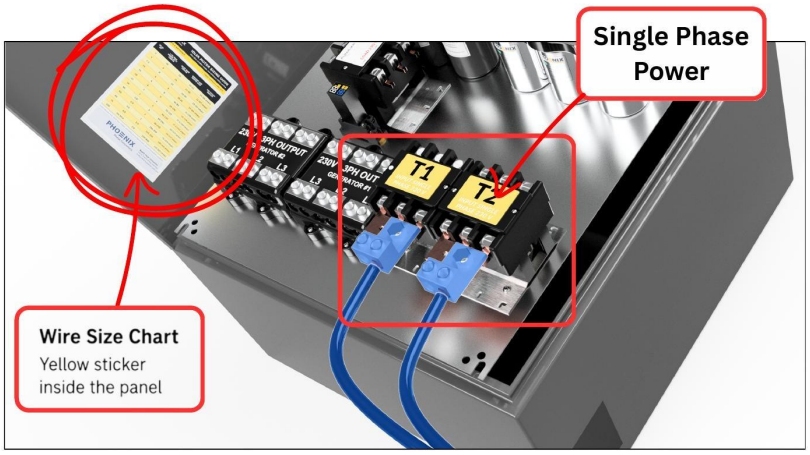
Single-phase input connection

PHOENIX PHASE CONVERTERS

Model Specs

Dual Zone 230V Series - SINGLE PHASE CONNECTION

Dual Zone must be taught as a sequence, not just a spec table. Each zone needs labels, photos, startup order, and a support handoff that shows both idlers and the panel connections.



Single-Phase Line Connection

Connect the incoming single-phase power line to terminals **T1** and **T2**.

Use **bolt-down lugs** and properly sized wire for the installation. Refer to the wire size chart for the correct wire gauge. The wire size chart is based on a maximum wire run of **50 feet**.

It is recommended to keep the single-phase line as short as possible. If the wire run exceeds **50 feet**, increase the wire size by **one gauge size for every additional 50 feet** of distance.

All wiring should be installed in accordance with applicable electrical codes and performed by a qualified electrician.

IMPORTANT: Verify utility transformer size, service capacity, breaker/fuse rating, and wire size before installation. Undersized supply equipment may cause low voltage, overheating, equipment damage, or failure to start. Installation must be performed by a qualified electrician.

Phoenix Phase Converters 800-417-6568 3

Land L1 and L2 from your 230V single-phase feeder on the input terminals. Bond the panel enclosure to your service ground. Size the feeder to the specified wire size for your DualZone model and distance.

Single-Phase Input Power for Smaller Units

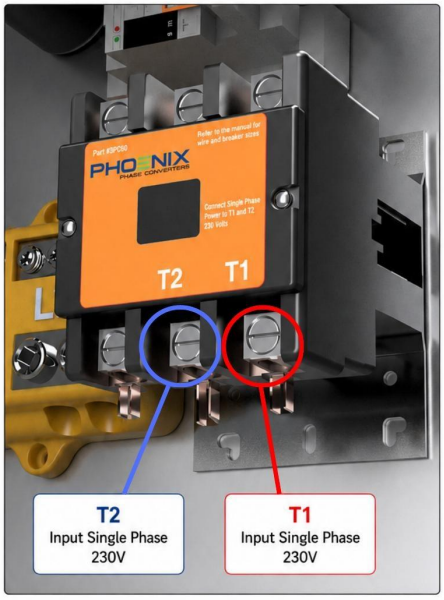
On smaller units, the incoming single-phase power connects to T1 and T2.

Connect the two incoming single-phase power wires to the terminals marked:

T1 Input Single Phase 230V

T2 Input Single Phase 230V

IMPORTANT:
On smaller units, connect incoming single-phase power only to **T1 and T2.**



L1 L2 L3

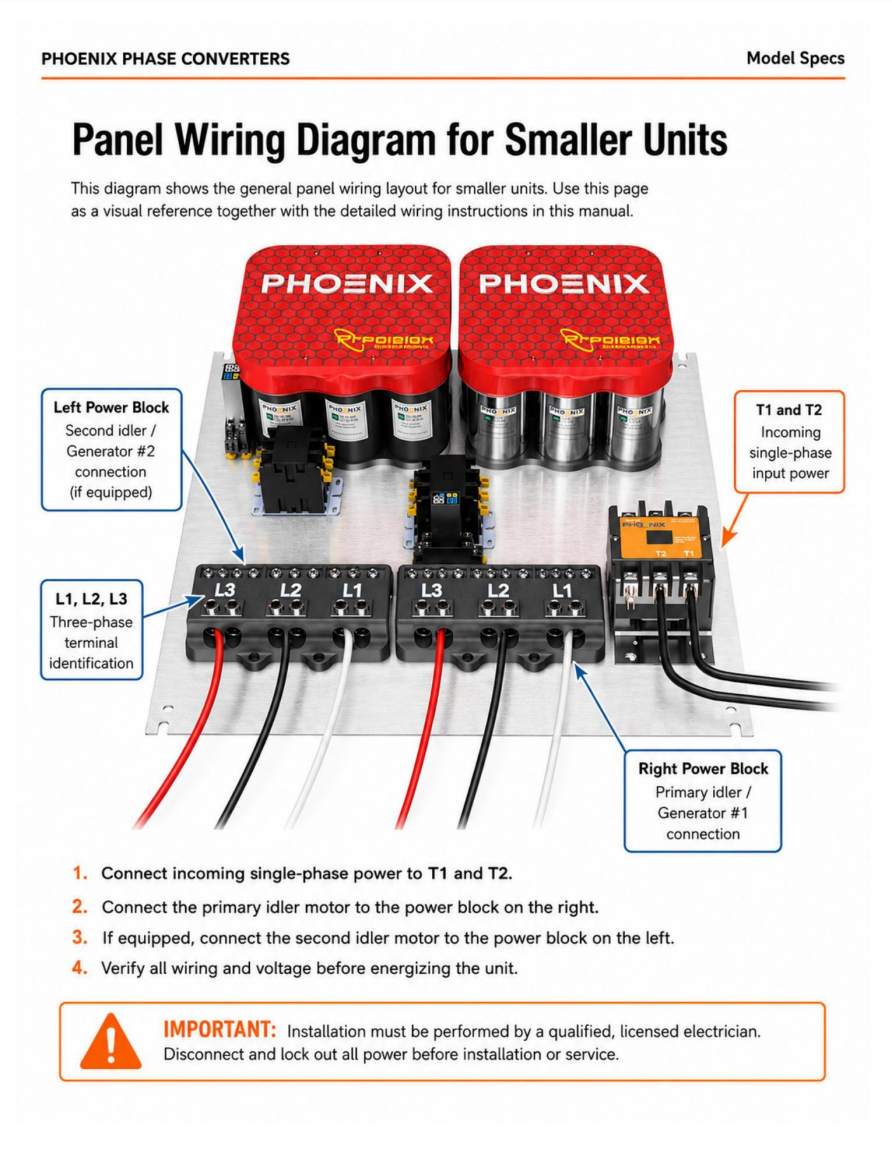
Do not connect incoming single-phase power to the generator output terminals.
Those terminals are for the three-phase output/load connections.

BEFORE ENERGIZING:

- Verify the input voltage is 230V single phase.
- Ensure all electrical connections are clean, secure, and properly tightened.
- Loose connections can cause overheating, voltage issues, or equipment damage.

For recommended wire and breaker sizes, refer to the wire size chart inside the panel or see Part #3PC60 in this manual.

Panel wiring diagram



PHOENIX PHASE CONVERTERS Model Specs

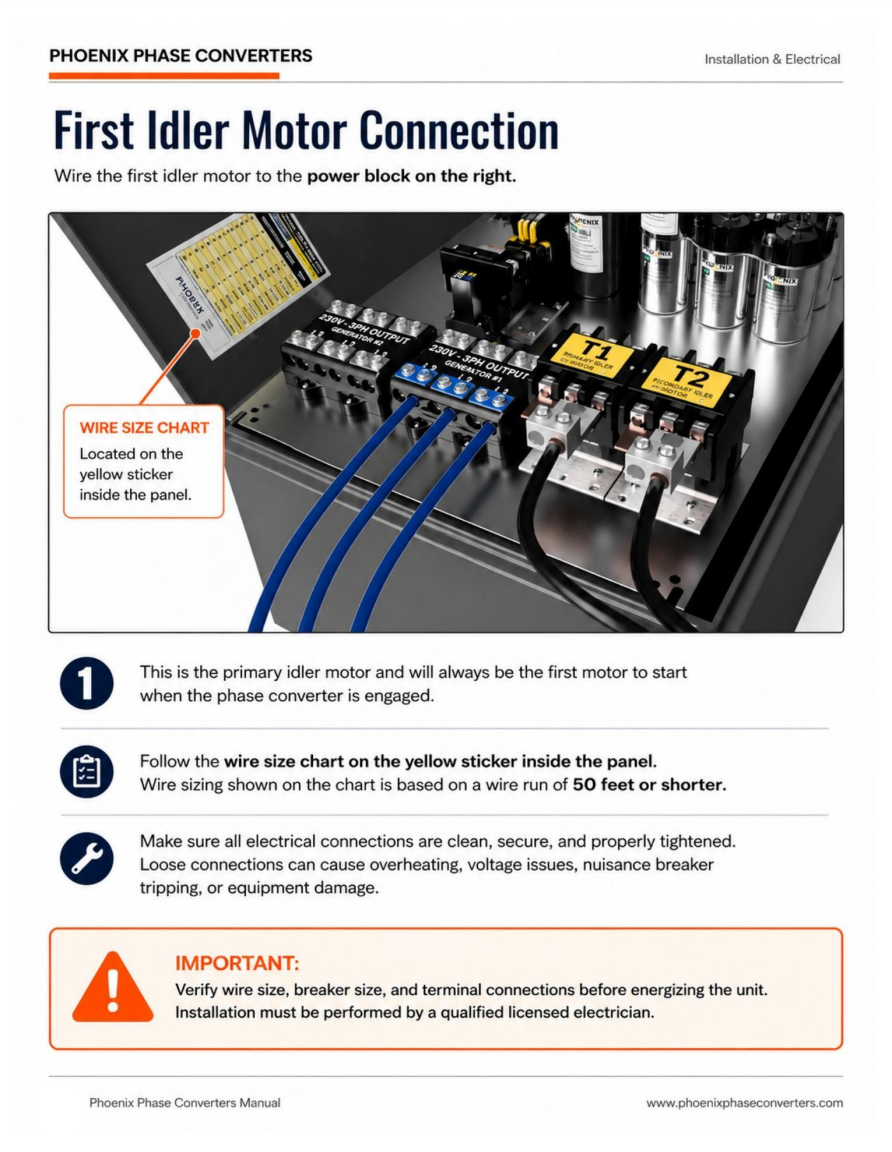
Panel Wiring Diagram for Smaller Units

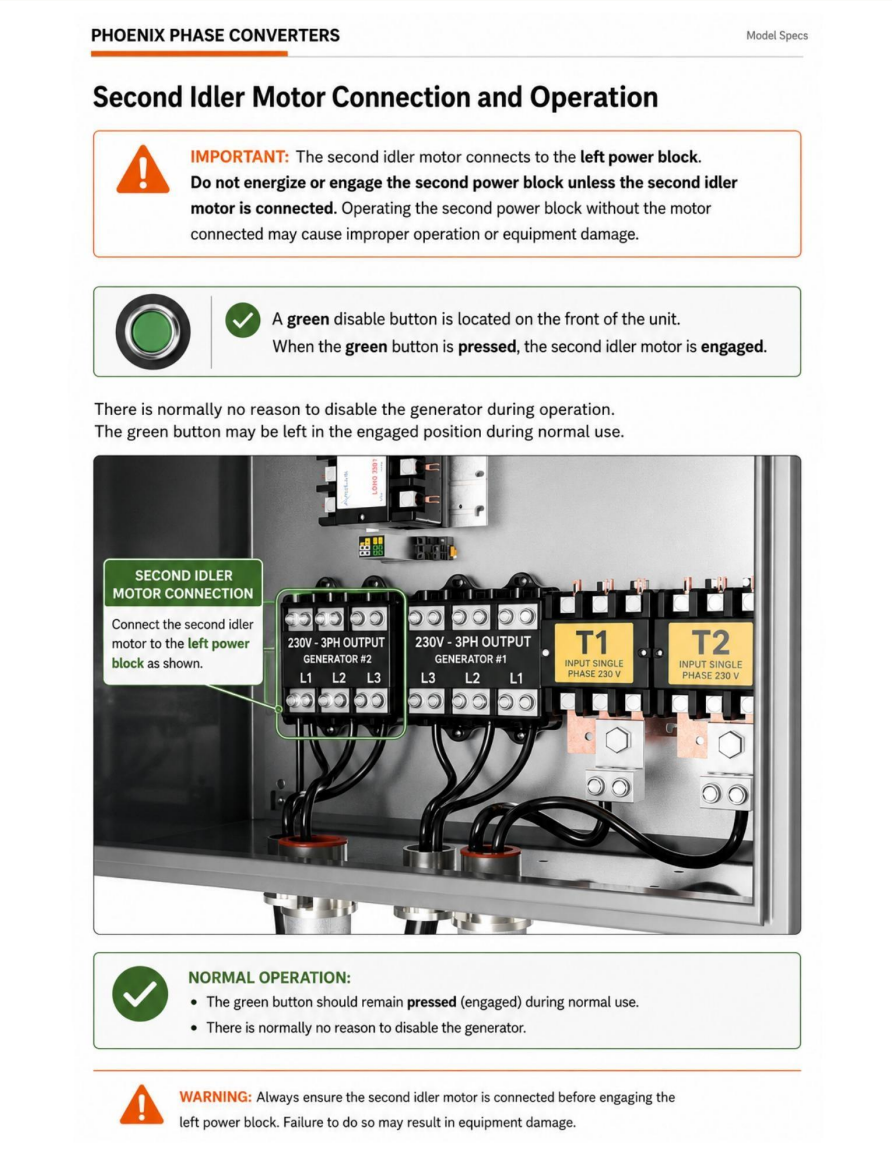
This diagram shows the general panel wiring layout for smaller units. Use this page as a visual reference together with the detailed wiring instructions in this manual.

1. Connect incoming single-phase power to T1 and T2.
2. Connect the primary idler motor to the power block on the right.
3. If equipped, connect the second idler motor to the power block on the left.
4. Verify all wiring and voltage before energizing the unit.

IMPORTANT: Installation must be performed by a qualified, licensed electrician. Disconnect and lock out all power before installation or service.

Idler motor connections





PHOENIX PHASE CONVERTERS

Model Specs

Second Idler Motor Connection and Operation

IMPORTANT: The second idler motor connects to the **left power block**. **Do not energize or engage the second power block unless the second idler motor is connected.** Operating the second power block without the motor connected may cause improper operation or equipment damage.

A **green** disable button is located on the front of the unit.
When the **green** button is **pressed**, the second idler motor is **engaged**.

There is normally no reason to disable the generator during operation.
The green button may be left in the engaged position during normal use.

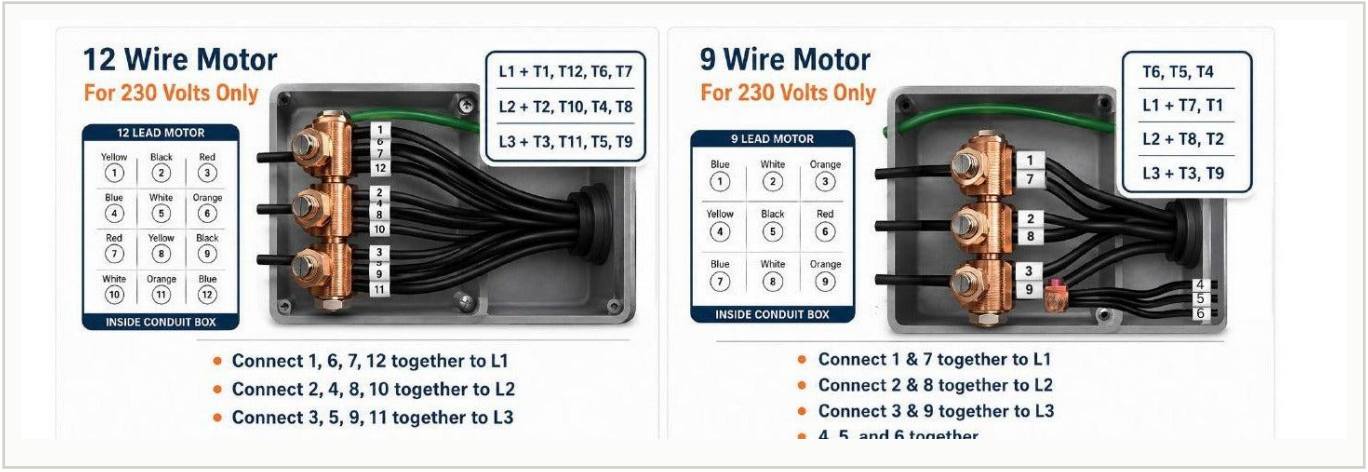
NORMAL OPERATION:

- The green button should remain **pressed** (engaged) during normal use.
- There is normally no reason to disable the generator.

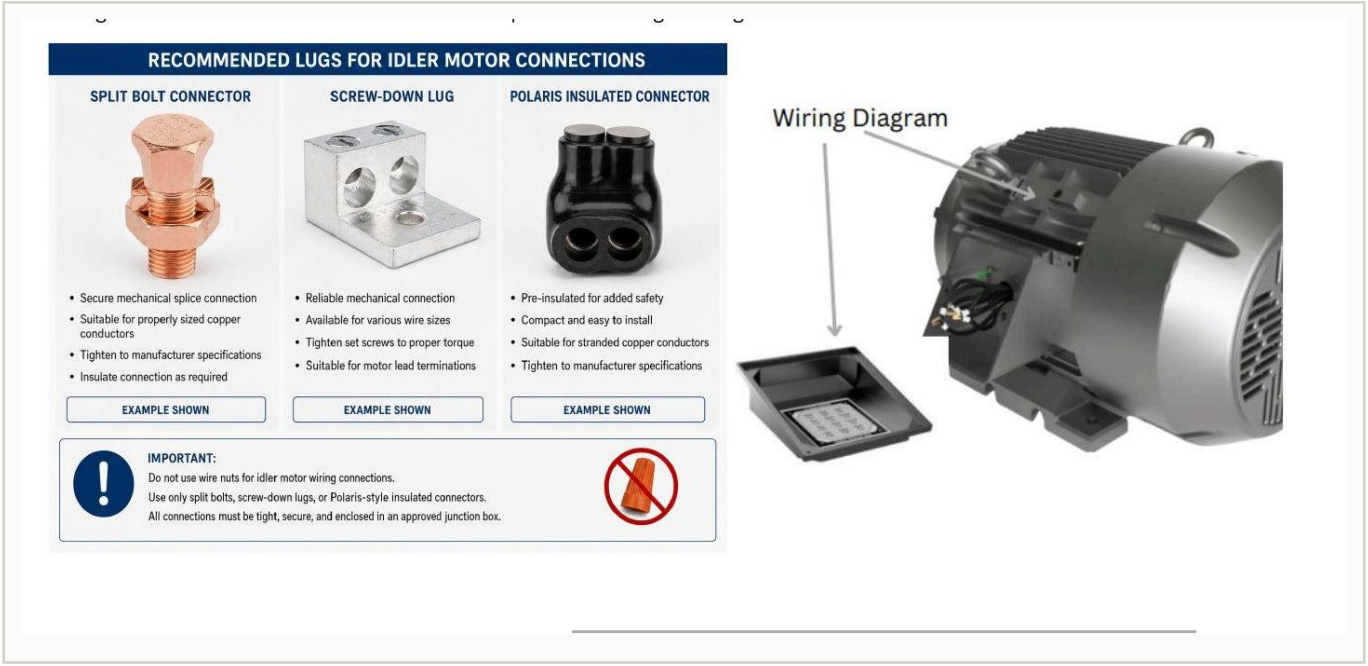
WARNING: Always ensure the second idler motor is connected before engaging the left power block. Failure to do so may result in equipment damage.

Each idler has its own dedicated terminal set inside the panel. Motor 1 handles the base load and starts first; Motor 2 auto-cycles in when the load exceeds the base-load threshold, giving the DualZone series its "Auto-Zoned" character.

Motor wiring — 9-lead and 12-lead



Lug types



Before testing


PHOENIX PHASE CONVERTERS

Model Specs

Before Testing

Before testing the phase converter, verify that the panel, conduit, and idler motors are installed correctly and that all wiring connections are complete, secure, and tightened properly. Confirm that the installation area is clear and safe before energizing the unit.

Before Testing Checklist

- Verify incoming single-phase power is connected correctly.
- Verify the idler motor wiring is complete and matches the motor wiring diagram.
- Verify all power connections are tight and secure.
- Confirm the motors are mounted securely and able to rotate freely.
- Make sure the area around the motors and panel is clear of tools and debris.
- Do not energize the unit until all covers, wiring, and connections have been checked.

This page is intended as a visual pre-start inspection reference.

IMPORTANT

Installation and testing must be performed by a qualified, licensed electrician.
Disconnect and lock out all power before making wiring changes or servicing the unit.

- All terminals torqued to spec.
- Panel door closed and sealed.
- No tools, wire scraps, or debris inside the panel.
- Grounding conductors verified.
- Downstream equipment disconnected.

DualZone - Ultimate Power Solution

Full DualZone product overview and installation walkthrough

youtube.com/watch?v=_1wJfM9_e3E

AutoStart · AutoLink · GPX Digital

> Three patented Phoenix features that solve real installation problems. Know what they do so you can spot when you need one.

AutoStart / AutoOff

The AutoStart feature (available on many NL/PL models as an add-on) makes the converter detect a load and start automatically, then shut off after 20 minutes with no load detected. This eliminates the "did I remember to turn off the converter?" problem and cuts standby energy waste.



AutoStart Phase Converter (Patented)
Overview of the patented AutoStart / AutoOff feature
youtube.com/watch?v=I7FB9yCpa_U



AutoStart / AutoOff V2 Technology
V2 evolution of the AutoStart / AutoOff feature
youtube.com/watch?v=tuj1e5iulfE

AutoLink

AutoLink is a patented Phoenix product that solves two problems: (1) pairing two phase converters together to double their combined output, and (2) preventing a compressor or transformer from re-energizing simultaneously with the converter after a power interruption.

When utility power is lost and then restored, the AutoLink box holds the downstream load off until the converter has fully spun up. Without AutoLink, a compressor or transformer trying to restart with the converter causes an overcurrent trip nearly every time.



AutoLink Box - Pairing Converters
How the AutoLink Box pairs two converters together
youtube.com/watch?v=g6pZ038j9Gg

AutoLink Box in Action - Real Customer
Real customer footage of AutoLink pairing two converters
youtube.com/watch?v=O6OGRZ7YeIs

Pairing to AutoLink - 3D Animation
3D animation of the AutoLink pairing sequence
youtube.com/watch?v=yjzqhyd5Oro

Phoenix AutoLink Overview
AutoLink product overview
youtube.com/watch?v=QcsCWEmQXBM

Digital AutoLink Introduction
Introduction to Digital AutoLink
youtube.com/watch?v=QbqVHpzWDnA

GPX Digital Series

GPX Digital converters take a rotary and add IGBT-based digital filtering, cloud-connected energy monitoring, WiFi remote on/off, 7-day scheduling, and a 24-hour timer. GPX is the cleanest power output in the Phoenix lineup and is the right choice for inverter-based TIG welders, CNC plasma tables, and any application where power quality is critical.

- Voltage output selectable: 230V and/or 460V three-phase (208V also available).
- WiFi/cloud monitoring: track KWh usage, voltage balance, and status from any device.
- 7-day scheduling: automatically turn on before the shop opens and off after close.
- Ultra-quiet digital operation compared to rotary.

Introducing the Digital GPX
Digital GPX product introduction
youtube.com/watch?v=1QptrhvgH4M



New Digital Converters

New digital converter product overview

youtube.com/watch?v=Qd3Cm3o0X-U

Preventive Maintenance Schedule

> Ten minutes per month saves thousands in emergency repairs. Every phase converter is a rotating machine with wear items — treat it like one.

Monthly (5 minutes)

- ☐ Visual inspection: no leaks, discoloration, or physical damage
- ☐ Listen for unusual noises during startup and running
- ☐ Verify all three voltage legs on the panel voltmeter — imbalance under 3 %
- ☐ Check the surrounding area is clear for airflow

Quarterly (15 minutes)

- ☐ Blow out dust and debris with compressed air (30 PSI max)
- ☐ Clean cooling fan blades at the idler non-drive end
- ☐ Torque check every accessible electrical connection
- ☐ Inspect capacitors for swelling or leakage
- ☐ Wipe down enclosure and check gasket integrity

Annually (30–60 minutes)

- ☐ Measure insulation resistance on all three output phases to ground with a megger — any reading below 1 MΩ indicates insulation breakdown
- ☐ Test every capacitor with a capacitance meter — replace any drifted more than 10 % from rated value
- ☐ Check bearing condition by spinning the idler by hand with power off
- ☐ Verify voltage balance under full load
- ☐ Torque every electrical connection to spec
- ☐ Update your maintenance log with date, findings, and any parts replaced

Every 5 years or 20,000 hours

- ☐ Consider proactive bearing replacement — much cheaper than a burned-out stator
- ☐ Consider proactive replacement of run capacitors as a matched set
- ☐ Inspect contactor contacts for pitting; replace if heavily worn

Keep a maintenance log. A simple paper log inside the panel door with dates, findings, and any parts replaced is worth its weight in gold when you call support — we can pinpoint the failure trend from a five-year log in under two minutes.

When to Call Us — Escalation Guide

> Some problems are easy fixes. Some are dangerous. Some are covered under warranty. Here is how to know when to stop and call.

Stop and call immediately if…

- You smell burning insulation coming from the converter — this indicates winding damage that gets worse fast.
- You see smoke or scorch marks anywhere in the panel.
- You feel high heat on the panel enclosure from outside (over 140 °F / 60 °C).
- A capacitor has ruptured, leaked, or the pressure vent has blown off.
- The contactor is stuck closed — you cannot shut the converter off.
- Any exposed conductor is arcing or has visible burn damage.
- You are unsure whether a step in this manual is safe to perform. Electricity does not give second chances.

Call for warranty support if…

- Your unit is under warranty and has developed a defect that is not the result of installation error.
- A component (capacitor, contactor, bearing) has failed prematurely — under 2 years for capacitors, under 5 years for bearings.
- The unit was received damaged or with a manufacturing defect.

Call for tech support if…

- You have worked through the relevant troubleshooting section and the problem persists.
- You are unsure about sizing before ordering additional equipment.
- Voltage imbalance exceeds 10 % and basic troubleshooting has not resolved it.
- You have replaced components but the same problem returns.
- Your motor lead configuration is not standard 9-lead or 12-lead and you need help mapping it.

What to have ready when you call

1. Model number from the nameplate (photo works best).

2. Order number, if you have it.
3. Voltage readings: input L1–L2, output T1–T2, T2–T3, T1–T3 at no load and under load.
4. A description of the equipment being run and its HP rating.
5. What has been tried already.
6. A short video of the unit while the problem is occurring (30 seconds is plenty). Text it to **602-962-8859**.

> **Phoenix Phase Converters Support**
>
> **800-417-6568**
>
> Text a photo or video: **602-962-8859**
> Email: **appliedindustrialmotors@gmail.com**
> Alternate: **866-418-9060**
> Monday–Friday 7 AM–5 PM MST · Saturday 7 AM–12 PM MST

Video Library

> Every official Phoenix Phase Converters video, categorized and QR-coded. Scan any QR code with your phone camera to open the video directly on YouTube. All videos are on the official channel **@PhoenixConverters**.

Troubleshooting

Troubleshooting Overview V1
youtube.com/watch?v=wqbkEcqrl_0

Troubleshooting Walkthrough
youtube.com/watch?v=bxn4VsamebQ

Voltage Drop Diagnosis

Checking for Voltage Drop
youtube.com/watch?v=gCGoN_nBVZ0

Checking for Voltage Drop V1.0
youtube.com/watch?v=ZszESYXSSwE

Technical Support

Technical Support - Wiring Help & Setup
youtube.com/watch?v=kRVblowXkcQ

Installation

Installation Video - Rotary Phase Converter
youtube.com/watch?v=VxW43xdWeIM

Installation Video - Push Button Models
youtube.com/watch?v=2lkahdPZSUY

Larger Models Installation - Audio
youtube.com/watch?v=1bx6URpngwM

Motor Wiring

9 Wire Leads Motor - Low Voltage 230V
youtube.com/watch?v=mVtPR4ESg6s

9 Wire Leads Motor (alt)
youtube.com/watch?v=UtUEzTf_tI8

12 Lead Motor - Low Voltage 230V
youtube.com/watch?v=BujkVL6-fG4

How to Connect a 12-Wire Motor
youtube.com/watch?v=BJ4zstQH-B8

AutoStart / AutoOff

AutoStart Phase Converter (Patented)
youtube.com/watch?v=I7FB9yCpa_U

AutoStart / AutoOff V2 Technology
youtube.com/watch?v=tuj1e5iulfE

AutoLink

AutoLink Box - Pairing Converters
youtube.com/watch?v=g6pZ038j9Gg

AutoLink Box in Action - Real Customer
youtube.com/watch?v=O6OGRZ7YeIs

Pairing to AutoLink - 3D Animation
youtube.com/watch?v=yjzqhyd5Oro

Phoenix AutoLink Overview
youtube.com/watch?v=QcsCWEmQXBM

Digital AutoLink Introduction
youtube.com/watch?v=QbqVHpzWDnA

GPX Digital

Introducing the Digital GPX
youtube.com/watch?v=1QptrhvgH4M

New Digital Converters
youtube.com/watch?v=Qd3Cm3o0X-U

DualZone

DualZone - Ultimate Power Solution
youtube.com/watch?v=_1wJfM9_e3E

Engineering & Behind the Scenes

Inside the Build - Rotary Engineering
youtube.com/watch?v=zXhlukOD3xk

Product Series Overview

PL 2021 Series
youtube.com/watch?v=oKz9H3hdNJU

2022 Series Overview
youtube.com/watch?v=_2Gl4-yDwWY

2021 Series
youtube.com/watch?v=X9vS0nZTjPY

CNC Applications

CNC Precision Balanced Voltage
youtube.com/watch?v=fU0sDAQTuVY

Real-World Test - 3 CNC Machines
youtube.com/watch?v=zi9L7Jf3P8o

Multi-Application

Rotary & Digital for CNC/ HVAC/Welding
youtube.com/watch?v=9ZImw4W0EWk

Full Video Channel

Phoenix Phase Converters Channel
youtube.com/watch?v=@PhoenixConverters

Full Model Reference Tables

> Every Phoenix rotary phase converter sizing table, in one place.

Load-Type Sizing Table (all NL / PL / DualZone models)

Use this table to select the converter HP for your equipment. Cross-reference your equipment HP with the load type column.

Load type	Multiplier	Equipment examples
Easy load	×1.0	Drill press, milling machine, saw motors
Medium load	×1.5	Mixer, disc sander, band saw, meat grinder, compactor, hoist
Hard load	×2.0	Lathe, wide belt sander, brake, shear, hoist, hydraulic pump, pumps, granite saw, CNC
Very hard load	×2.5	Air compressor, blower, vacuums, cryo chamber, HVAC
CNC spindle	See table	Spindle motor and all other CNC motors

Model Sizing Table (NL / PL Series, 3–150 HP)

Model	Easy	Medium	Hard	Very Hard	Resistive Amps	KVA	CNC Spindle
GP3	2.5	2	1.5	1	4	1.6	1.5
GP5	3	3	2	1.5	7	2.9	2.5
GP7	7	5	3	2.5	11	4.5	3.5
GP10	8	6	5	3	15	6	5
GP15	10	8	7.5	5	20	8	7.5
GP20	15	12	10	7.5	25	11	10
GP25	20	15	12	10	30	13	10
GP30	25	20	15	10	37	15	15
GP40	30	25	20	15	50	21	20
GP50	40	30	25	20	61	25	25

Model	Easy	Medium	Hard	Very Hard	Resistive Amps	KVA	CNC Spindle
GP60	50	40	30	25	72	30	30
GP75	60	50	40	30	100	29.6	35
GP100	75	60	50	40	122	37.3	50
GP120	100	75	60	50	148	60	60
GP150	125	100	70	60	162	72	70

Electrical Specifications by Model (NL Series)

Model	HP	KW	KVA	Breaker Min	Breaker Max	Wire Size	Panel
GP5NL	5	3.3	4.43	10	30	8 AWG	20×12×8
GP7NL	7	5.2	6.4	15	40	6 AWG	20×12×8
GP10NL	10	7.5	13.01	20	60	6 AWG	20×12×8
GP15NL	15	11.2	12.79	30	80	6 AWG	20×12×8
GP20NL	20	14.9	17.35	40	100	4 AWG	20×12×10
GP25NL	25	18.6	21.7	50	125	4 AWG	20×12×10
GP30NL	30	22.4	26	60	150	2 AWG	20×12×10
GP40NL	40	29.8	43.37	80	200	1/0 AWG	20×12×10
GP50NL	50	37.3	54	100	250	2/0 AWG	24×12×16
GP60NL	60	44.7	65	125	300	3/0 AWG	24×12×16
GP75NL	75	55.9	81	150	350	4/0 AWG	26×12×20
GP100NL	100	74.6	87.21	150	400	300MCM	26×12×26

DualZone Auto-Zoned Models

Model	Base HP	Combined HP	Application
GP20/40NL	20	40	Small multi-machine shops
GP25/50NL	25	50	Medium multi-machine shops
GP30/60NL	30	60	Full CNC shop or large compressor

Model	Base HP	Combined HP	Application
GP40/80NL	40	80	Heavy industrial multi-machine
GP50/100NL	50	100	Industrial facility, 5+ machines

Warranty, Contact & Support Hours

Warranty coverage

- **Lifetime warranty** on the idler motor and cast-iron frame.
- **5-year warranty** on electrical panel components (capacitors, contactors, relays).
- **1 year of free phone support** from date of purchase — Glen and Danny answer the phone personally.
- Warranty is valid for original purchaser only; keep your original invoice.
- Damage from improper installation, water ingress on indoor-only units, or use outside rated capacity is not covered.

Contact Phoenix Phase Converters

Sales & Support: 800-417-6568

Alternate: 866-418-9060

Text (photos/videos): 602-962-8859

Email: appliedindustrialmotors@gmail.com

Web: phoenixphaseconverters.com

Hub / Resources: hub.phoenixphaseconverters.com

Customer Portal: app.phoenixphaseconverters.com

Support hours (MST)

- Monday – Friday: 7 AM – 5 PM
- Saturday: 7 AM – 12 PM
- Sunday: emergency text only

Company info

Phoenix Phase Converters (Applied Industrial Motors LC)
Family-owned since 2003

Over 40,000 units shipped
Proudly made in Phoenix, Arizona · USA